



THE INSTITUTE OF
MEDICAL SCIENCE,
THE UNIVERSITY OF TOKYO



**Centro de Investigación
en Alimentación y Desarrollo**

Single-cell RNA-seq data analysis

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Lecture roadmap

50 minutes of content, followed by 10 minutes for questions

01

Bulk RNA-seq context

A short introduction to reads and count matrices

8 min

02

Single-cell workflow

Preprocessing, clustering, markers, and annotation

28 min

03

Multiple samples

Batch effects, integration, and validation

8 min

04

Spatial transcriptomics

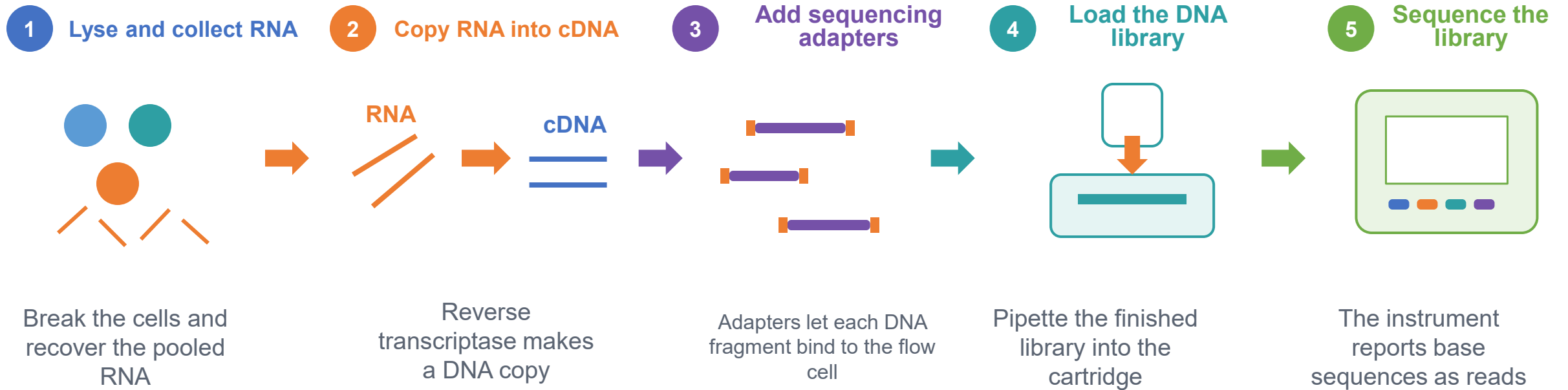
Spots, SCTransform, and tissue maps

6 min

RNA sequencing workflow

RNA becomes an adapter-tagged DNA library before it enters a short-read sequencer

Why make cDNA? The sequencer chemistry copies and detects adapter-bound DNA. cDNA is stable, can be amplified, and accepts the required adapters.



More RNA molecules produce more cDNA fragments and therefore more sequencing reads.

What the sequencer reads

- A read is a short piece of sequence, between 150 - 300 letters (bases) long.
- One sample gives millions of reads.
- Each read came from one RNA molecule.

GATTACAGGCATTCCAGTACCGA

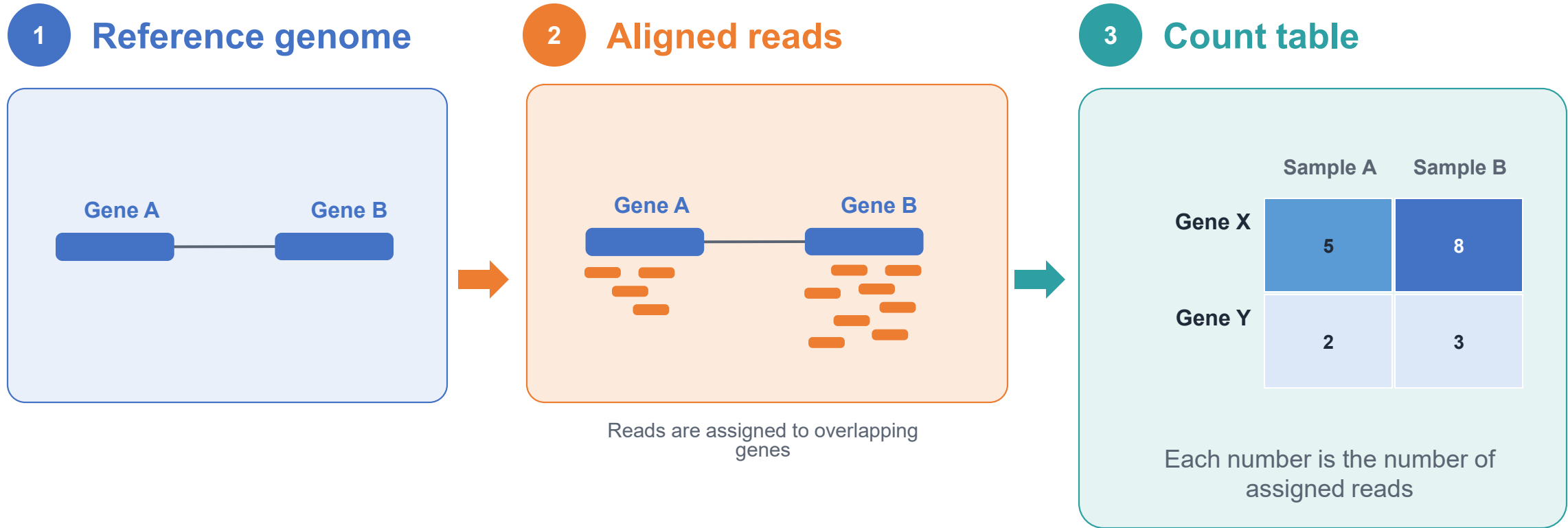
one read

The computer then looks for the place in the genome where each read fits (alignment).

That place tells us which gene the read came from.

From reads to gene counts

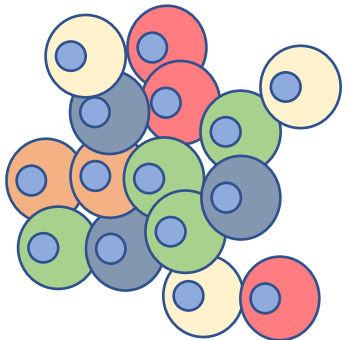
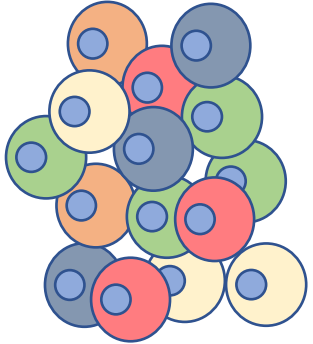
Alignment tells us which gene each sequenced fragment came from



Align → assign → count: the output is a gene-by-sample matrix.

RNA sequencing data (bulk)

Sample 1



Sample 2

	Gene 1	Gene 2
Sample 1	1	2
Sample 2	5	2

Bulk samples merge cells' information, masking each cell's identity

How 10x separates single-cell profiles

A cell barcode links every read back to the droplet in which it was captured

1 Partition

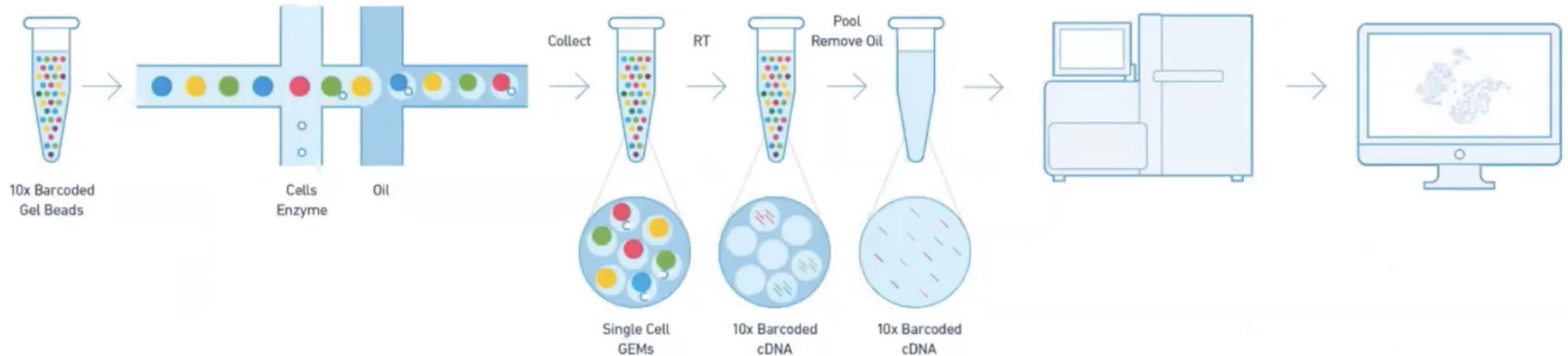
One cell + one barcoded bead enter a GEM

2 Capture

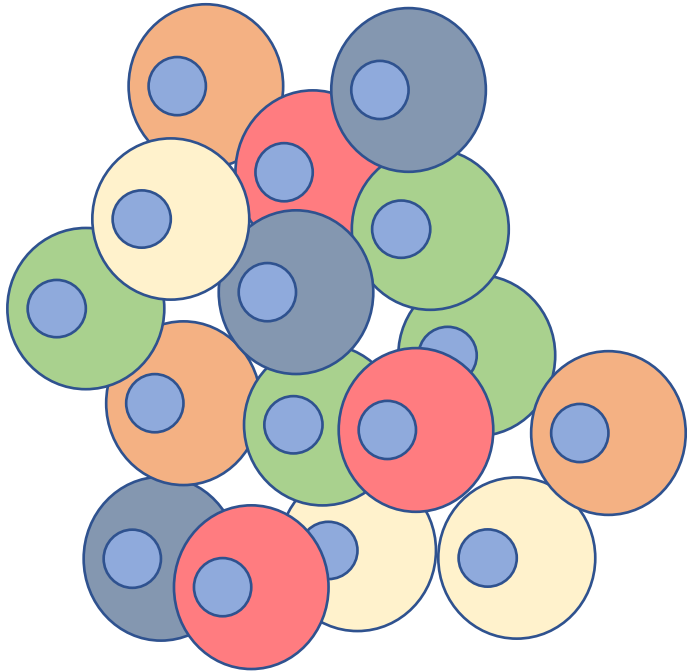
The cell lyses and RNA is copied into barcoded cDNA

3 Sequence

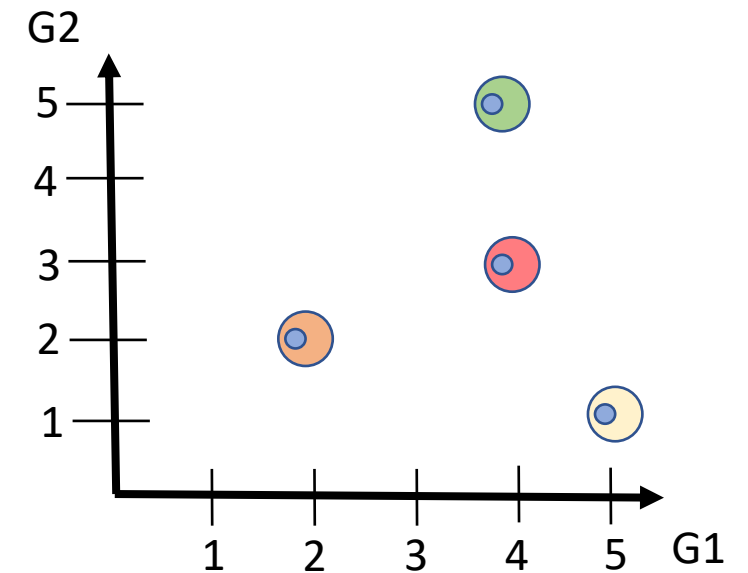
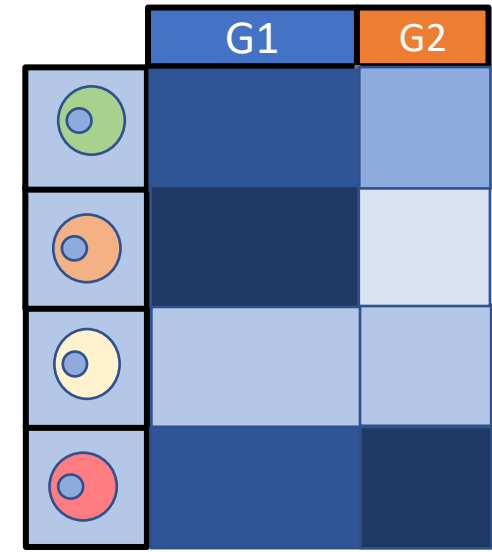
Pool libraries; cell barcodes recover identity



Single-cell RNA-seq data

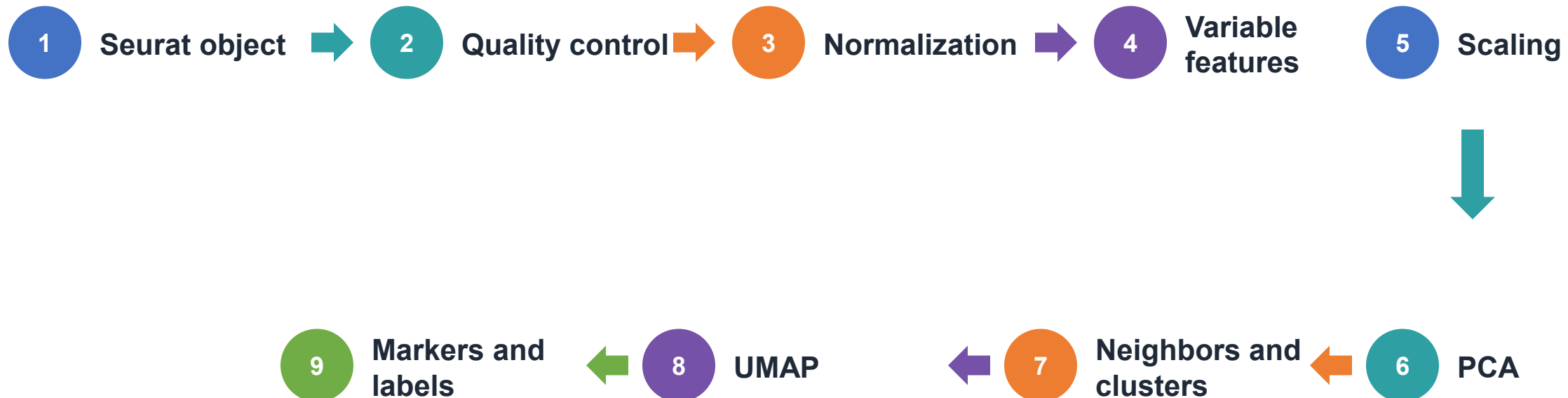


	G1	G2
	4	3
	5	1
	2	2
	4	5



Single-cell analysis workflow

The notebooks follow the same sequence from counts to biological labels



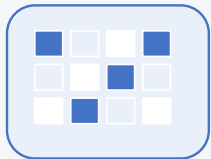
Each step reduces a technical obstacle or creates a representation needed by the next step.

At each step, we need to make a decision. There is no right answer or tool; each dataset should be explored.

The Seurat object

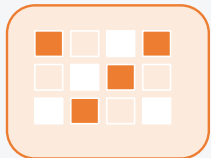
One object keeps the measurements, cell information, and analysis results aligned

RNA assay



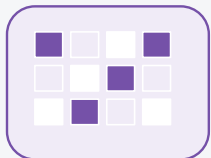
counts

raw integers



data

normalized values



scale.data

centered and scaled

1

Metadata

QC metrics, sample, cluster, and cell type

2

Reductions

PCA and UMAP coordinates for every cell

3

Graphs

Nearest-neighbor relationships used for clustering

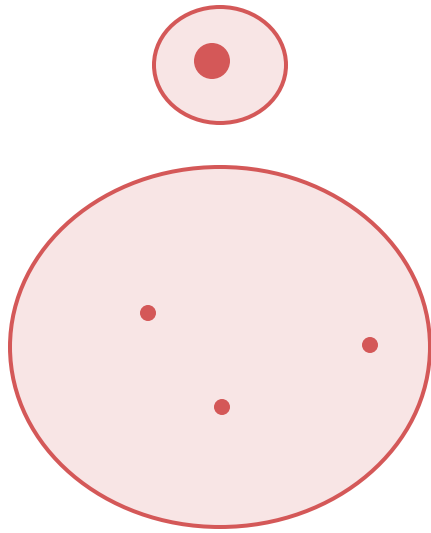
4

Identities

The current grouping used by plots and marker tests

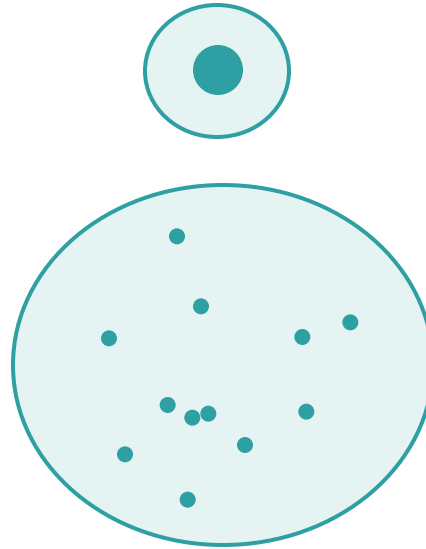
Quality control

Some droplets do not represent one healthy cell



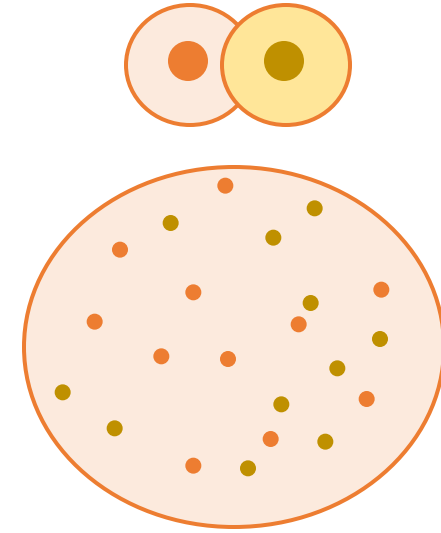
Low-quality or empty

Few genes
High mitochondrial fraction



Usable cell

Balanced counts
Many detected genes



Possible doublet

Too many counts
Too many detected genes

Goal: retain droplets that behave like single, intact cells before downstream analysis.

QC metrics

Each metric reveals a different failure mode

nFeature_RNA

genes detected

Low-quality or
empty

too few

Usable cell

usable range

Possible doublet

very high may indicate a doublet

nCount_RNA

total molecules

too few

usable range

very high may indicate a doublet

percent.mt

mitochondrial fraction

Expected high

inspect tissue-specific range

high may indicate a stressed cell

Thresholds come from the observed distributions and the biology of the sample.

Sequencing depth distorts raw counts

The reference gene is identical; only the number of sampled reads changes

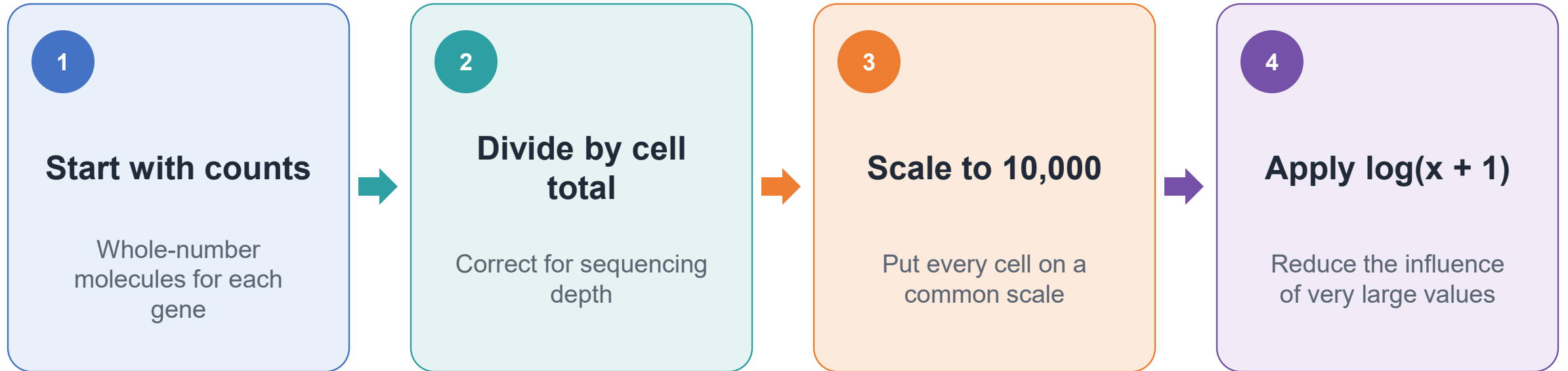


A deeper library samples more molecules from the same biology.

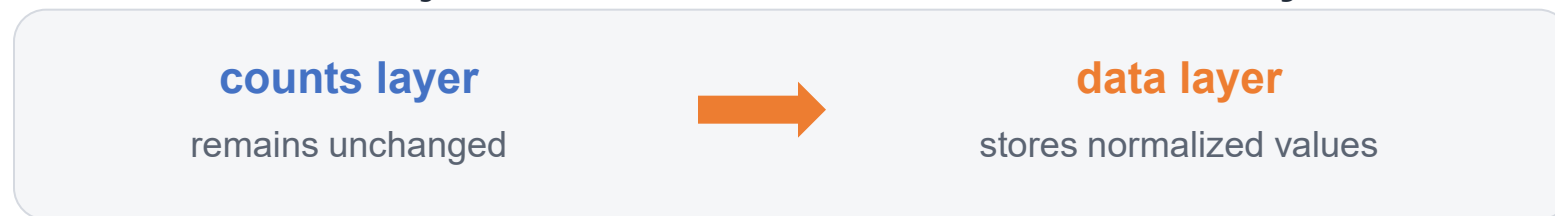
In addition, some genes might be more likely to be captured by the sequencing pipeline, for example, because of differences in GC content.

Log normalization

NormalizeData corrects depth and compresses very large values

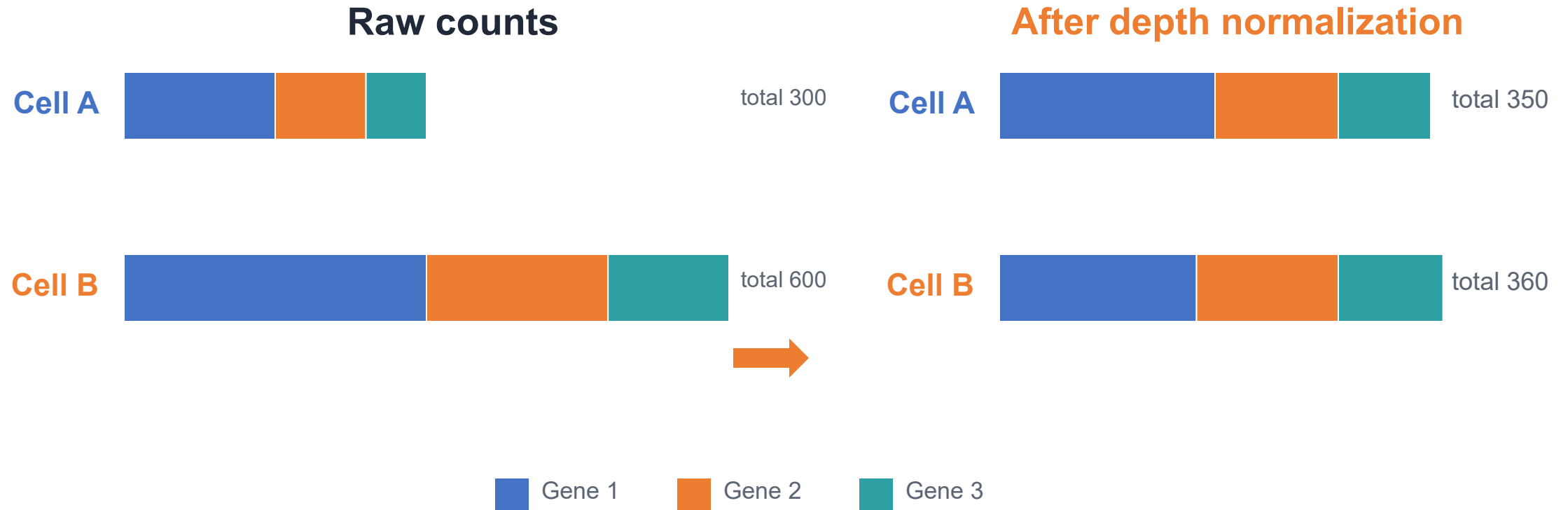


In the Seurat object, the data is stored in different layers



Effect of normalization

Depth is balanced without forcing cells to have identical expression profiles

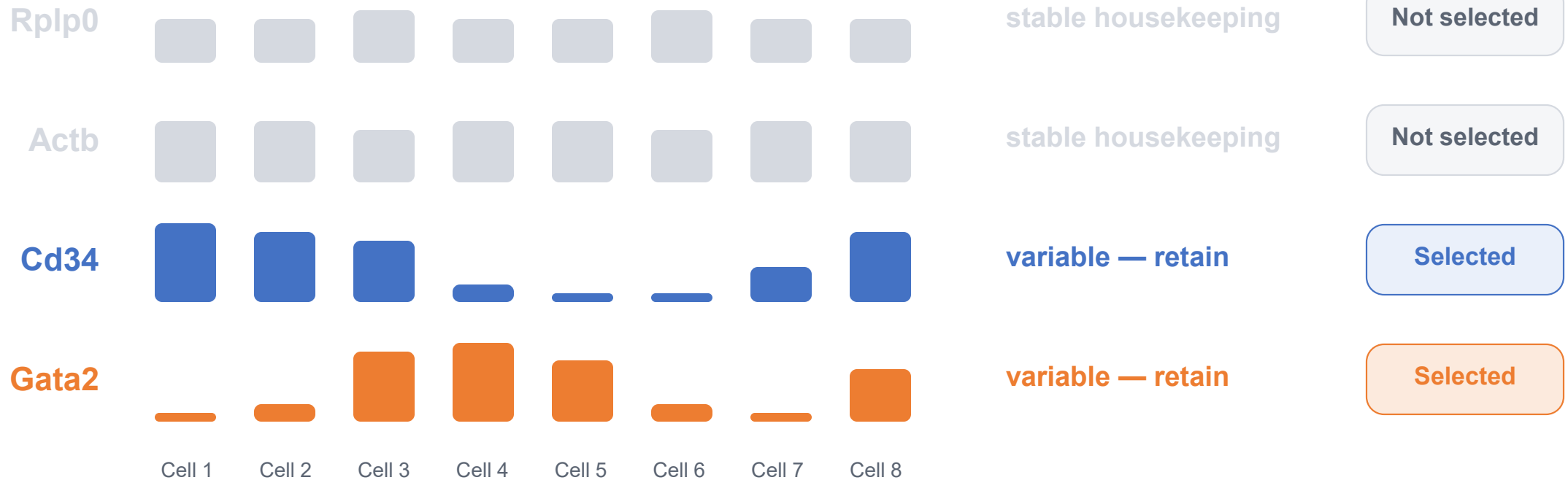


Comparable library sizes remove depth effects; unequal segment sizes aim to preserve biological differences.

Select variable features

Keep genes that change meaningfully across cells; de-emphasize nearly constant genes

Expression across eight cells



Stable genes add little information for separating cells. Variable genes carry the structure used by PCA and clustering.

Scaling makes genes comparable

For each selected gene: subtract its mean and divide by its standard deviation

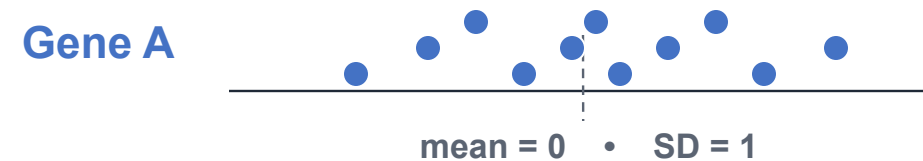
Before scaling



ScaleData



After scaling



In this workshop, scaled values feed into PCA and heatmaps. Most expression plots use the log-normalized values.

Why dimensionality reduction is necessary

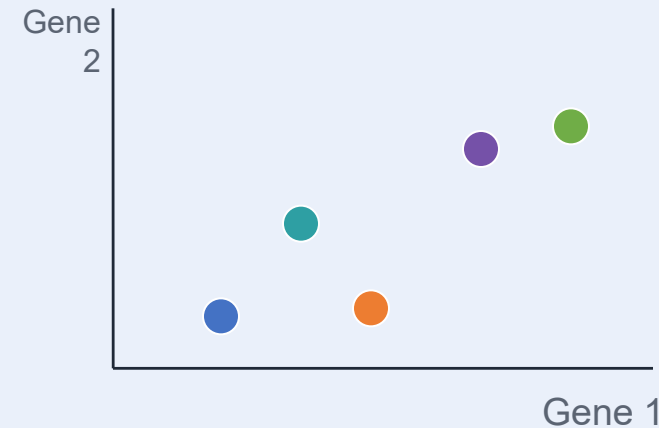
Each gene is one dimension; a cell has thousands of genes (dimensions)

	Gene 1	Gene 2	Gene 3	...
Cell 1	8	1	4	6
Cell 2	7	2	4	5
Cell 3	1	8	3	2
Cell 4	2	9	2	3

cells × genes



A cell is a point in gene space

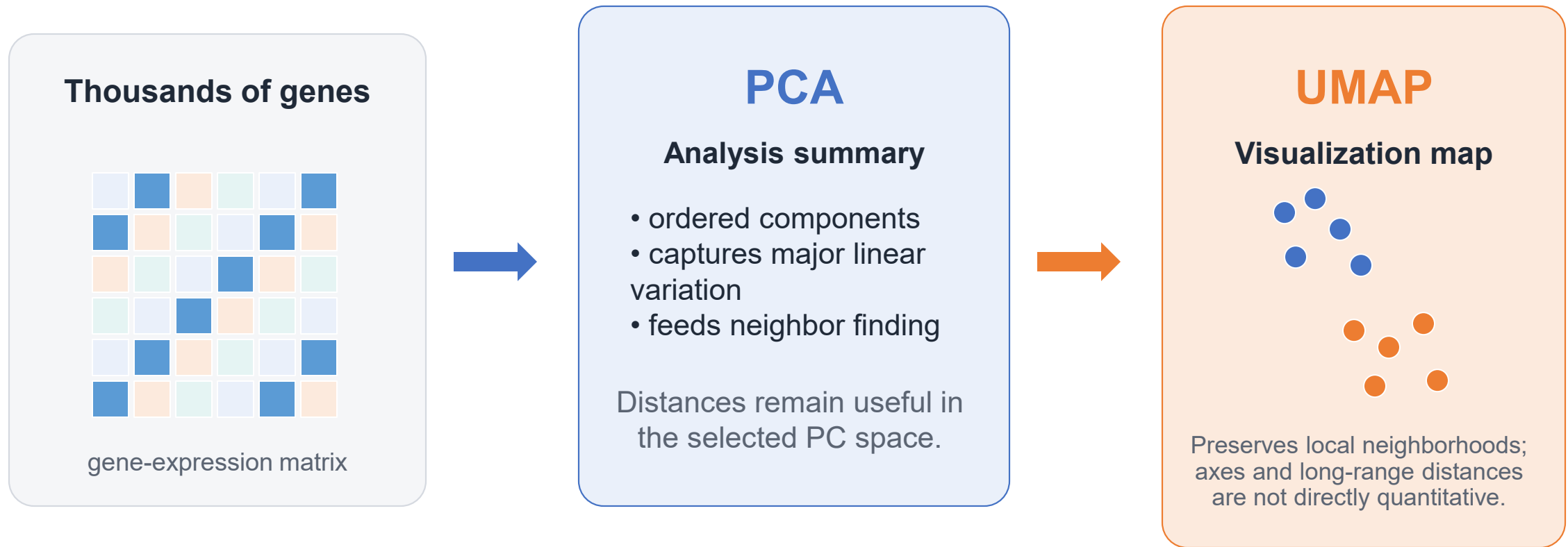


A third gene needs a third axis; thousands of genes cannot be viewed directly.

We reduce thousands of gene dimensions to a smaller representation that we can analyze and visualize.

Two roles for dimensionality reduction

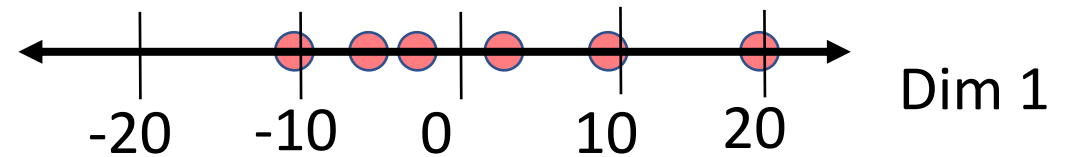
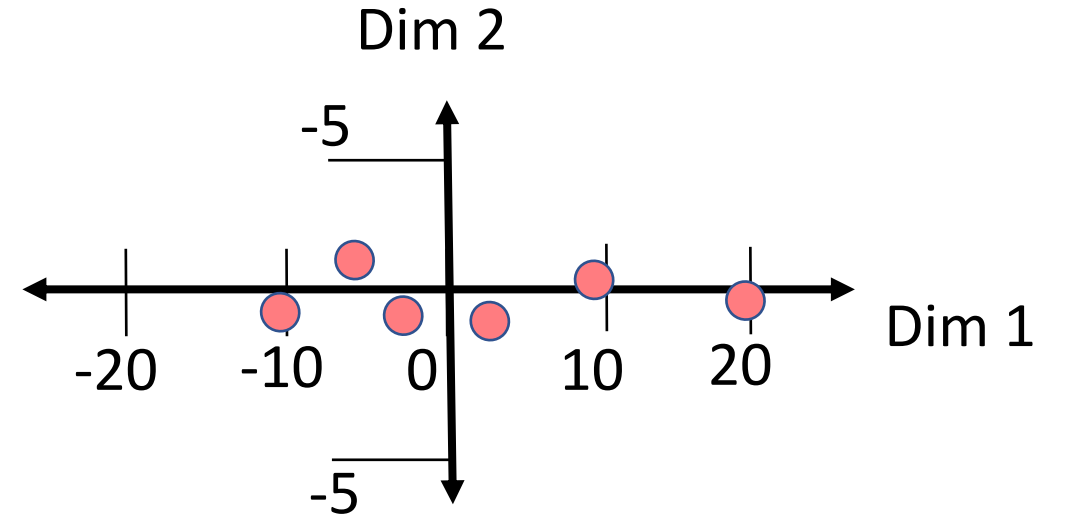
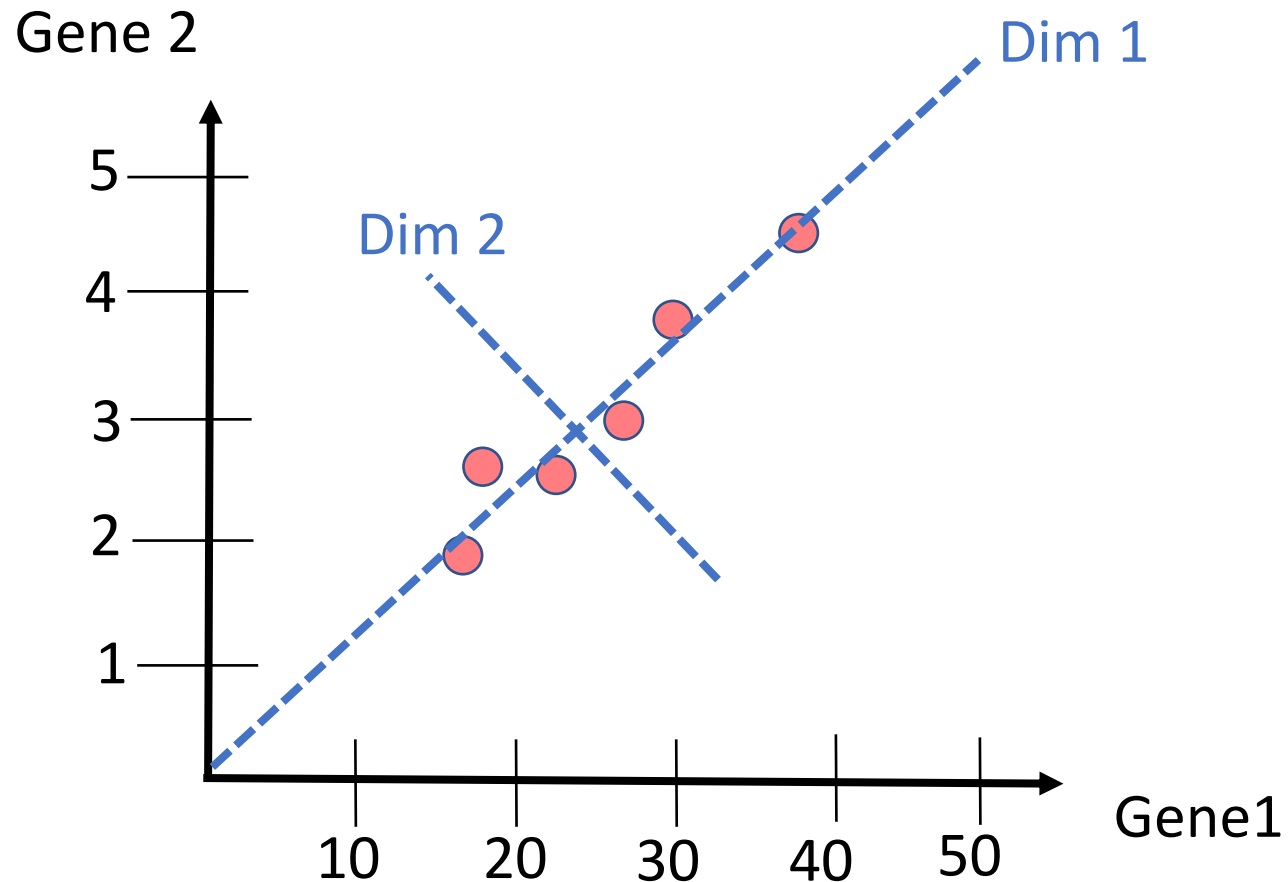
A compact analysis space and a readable visualization are related, but not interchangeable



Use PCA to summarize and compute; use UMAP to inspect the structure in two dimensions.

A PCA example

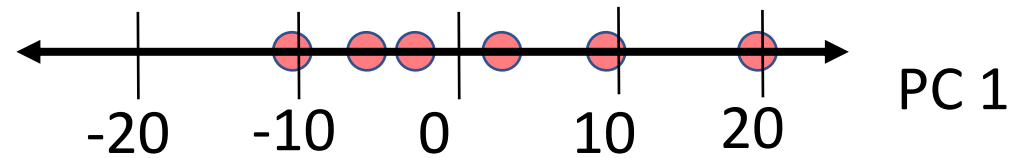
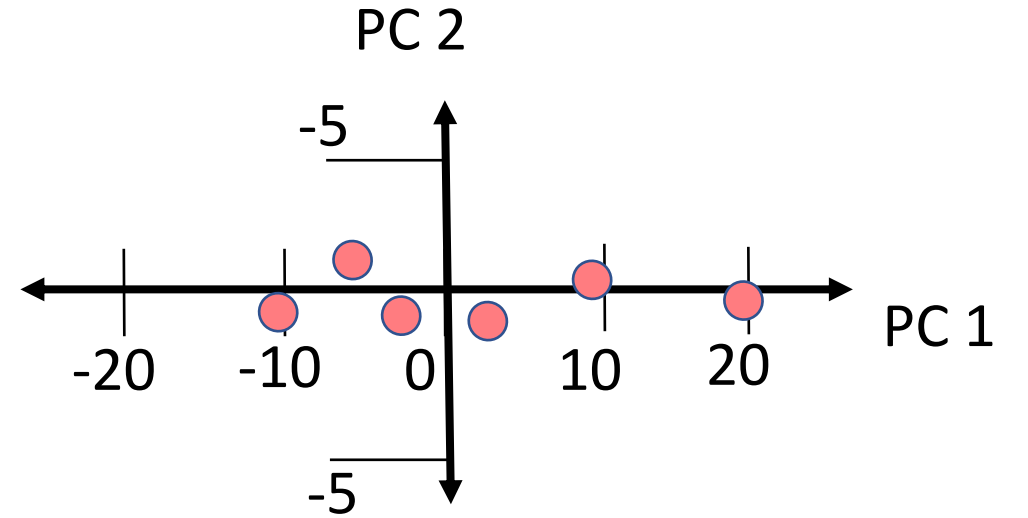
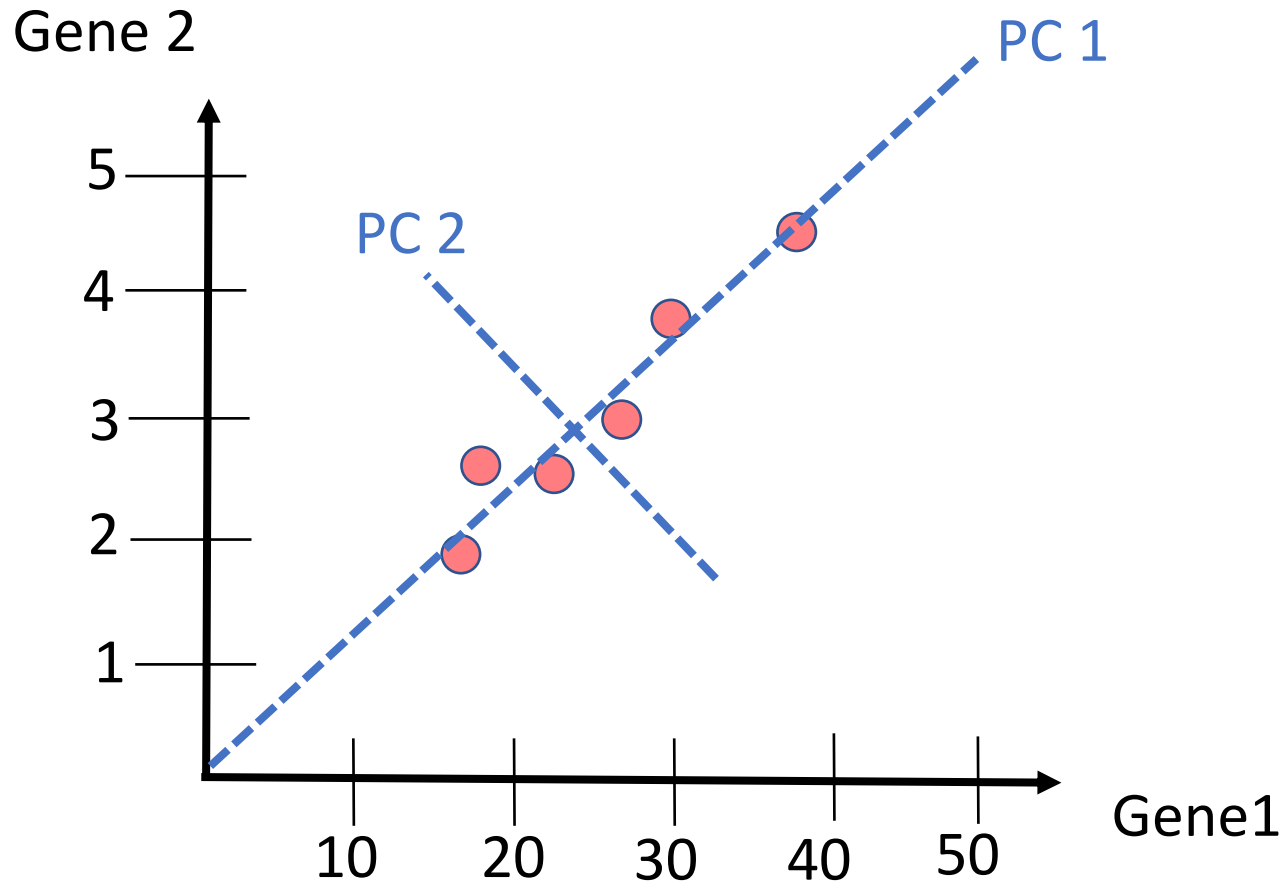
When two genes move together (correlated), one new axis can summarize most of their shared variation



We only need one dimension to describe the data

A PCA example

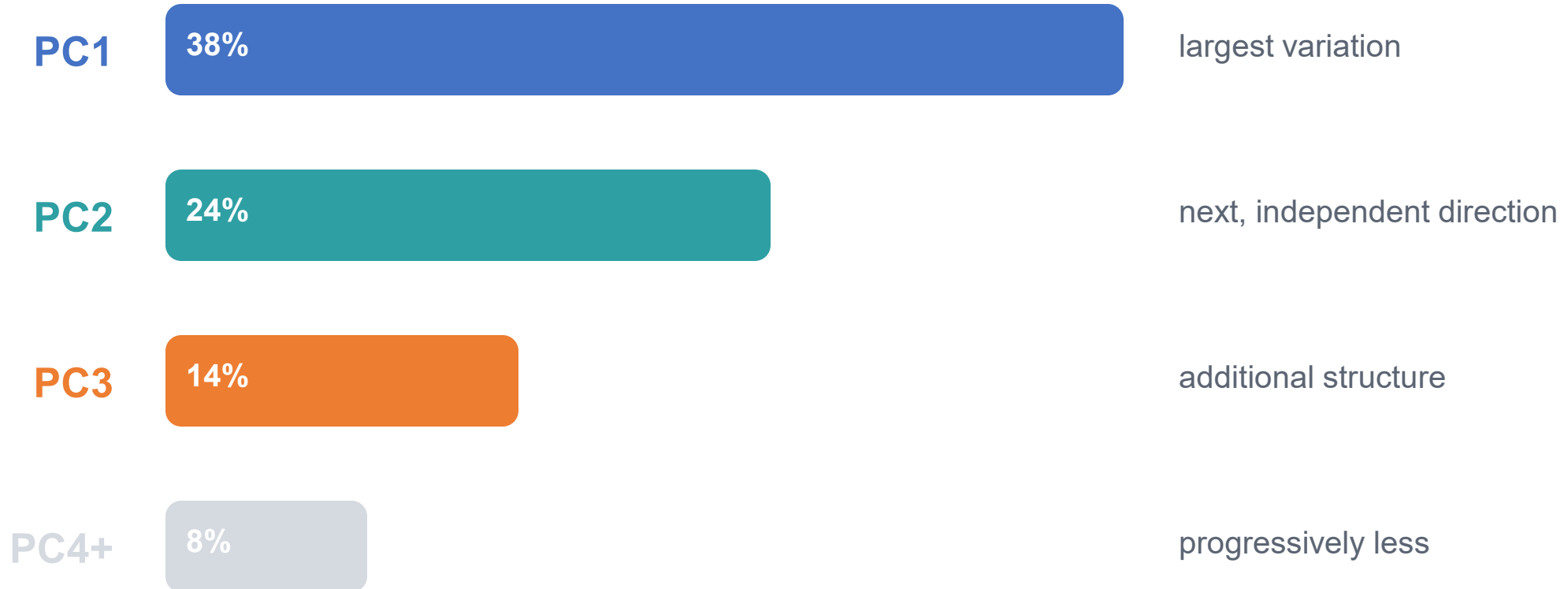
When two genes move together (correlated), one new axis can summarize most of their shared variation



We only need one dimension to describe the data

PCA creates ordered summaries

Each principal component combines many genes and captures a decreasing share of variation

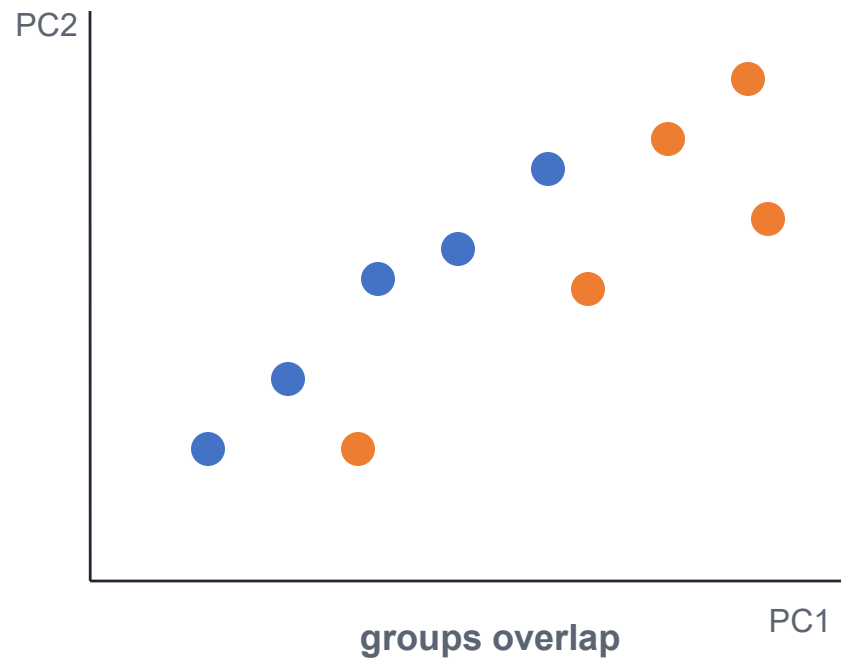


A PC is a weighted combination of genes. Cells receive one score on each PC.

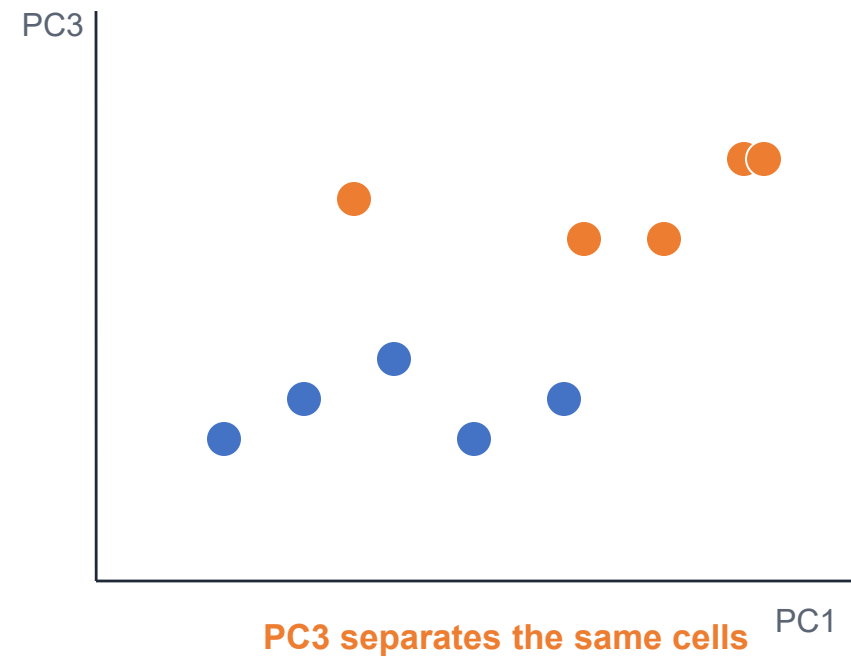
Important structure may appear beyond PC2

Two-dimensional PCA plots show only two of the many available components

PC1 vs PC2



PC1 vs PC3

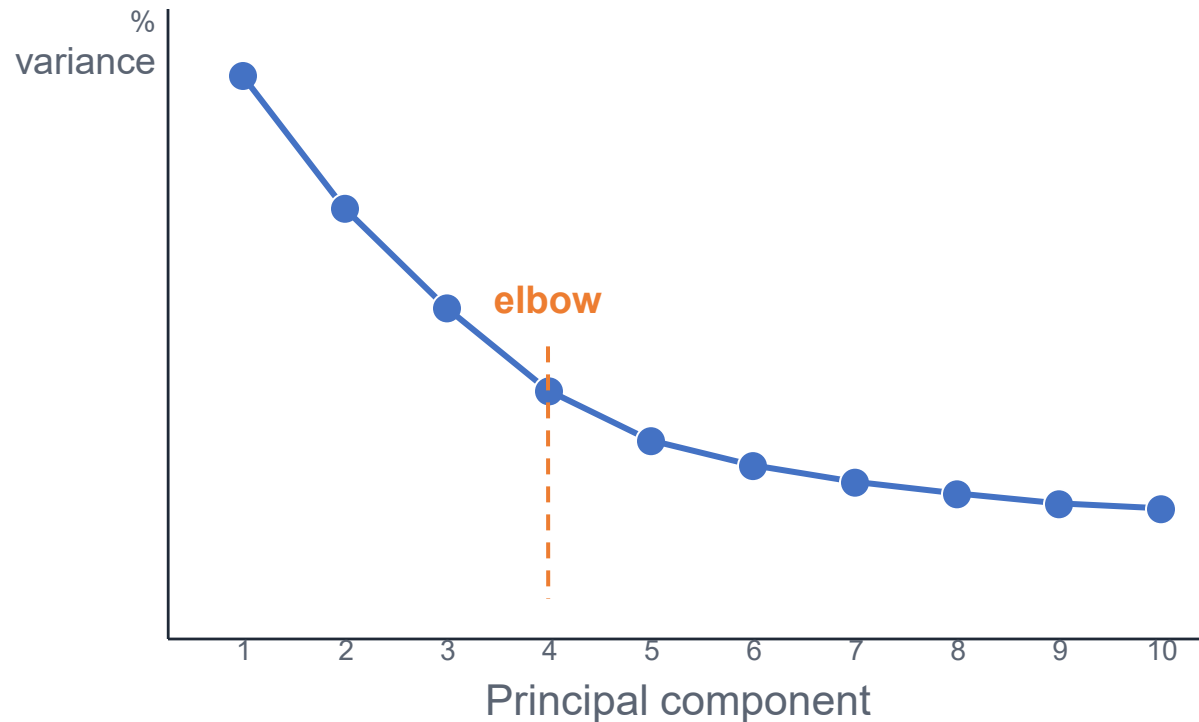


This is why downstream analysis uses several PCs, not only the two PCs we can draw.

How many PCs should we keep?

The elbow plot is a practical guide, not a rigid rule

Variance explained



Interpretation

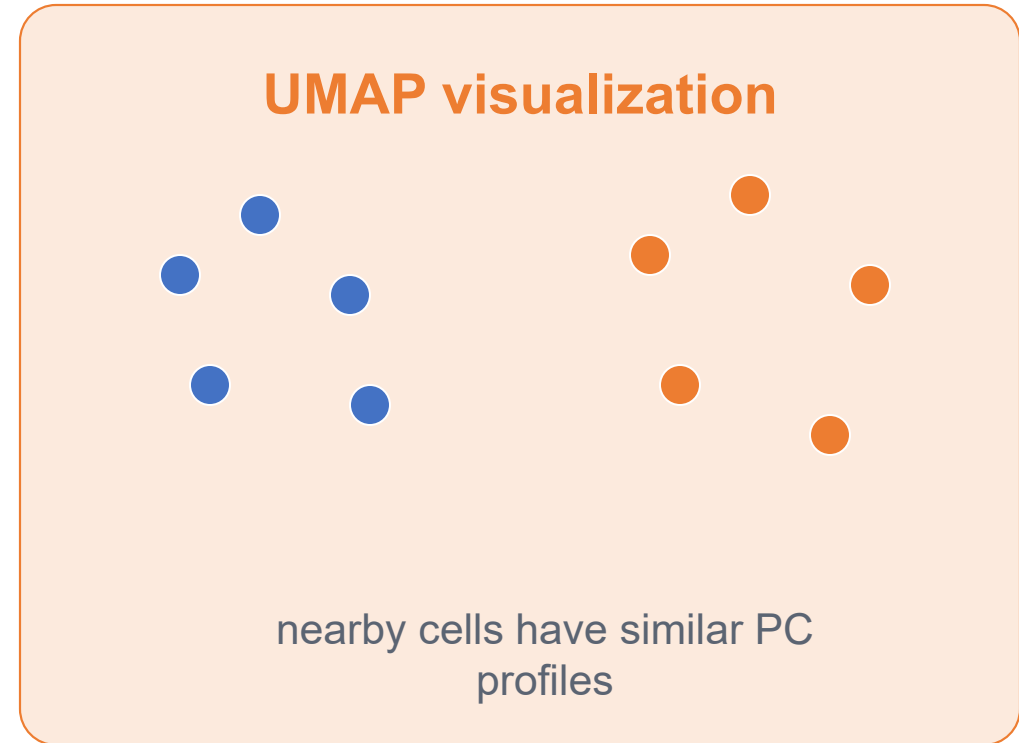
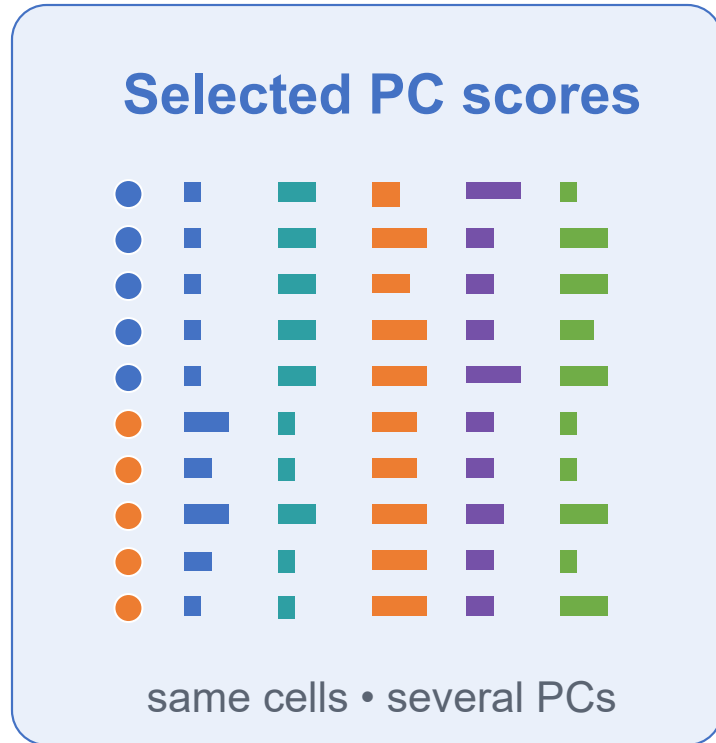
Keep components until the curve flattens.

Then check whether the chosen PCs preserve meaningful biological structure.

In general we would like to explore results using different number of PCs

UMAP turns selected PCs into a 2D map

It is a visualization of local neighborhoods, not another expression matrix



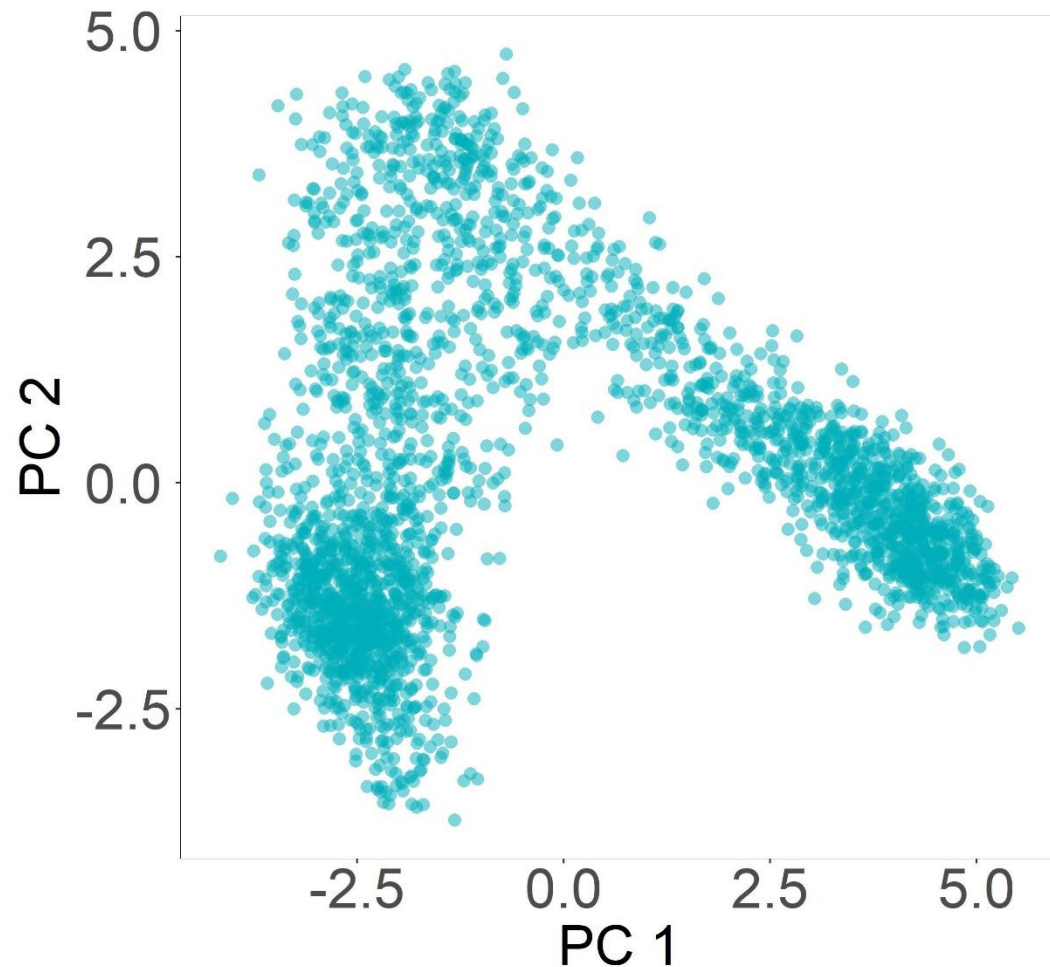
Use UMAP to view the data. The neighbor graph—not visual distance alone—defines the clusters.

Dimensionality Reduction

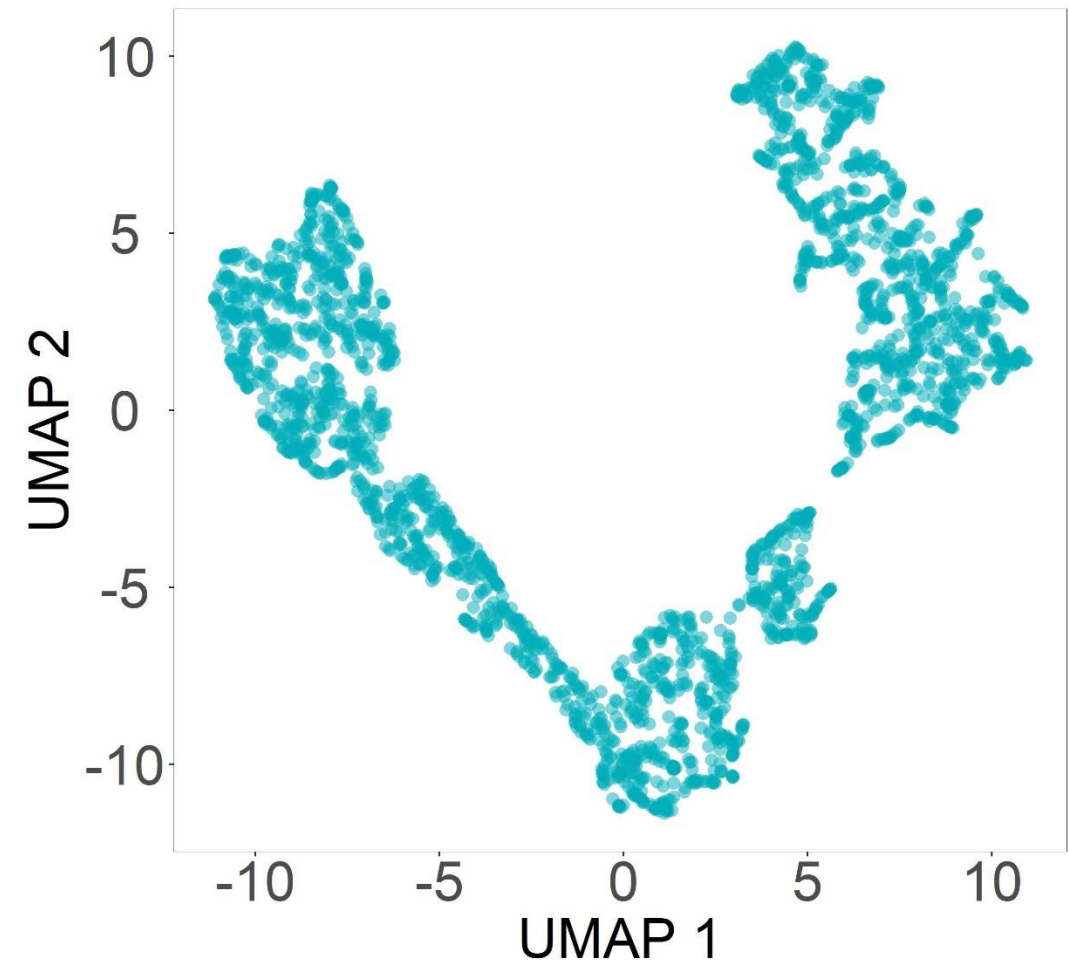
Hematopoietic Cells
2729 Cells, 3467 Genes

Paul, F. et al. Transcriptional heterogeneity and lineage commitment in myeloid progenitors. Cell 163, 1663–1677 (2015).

PCA



UMAP



Neighbors and clustering

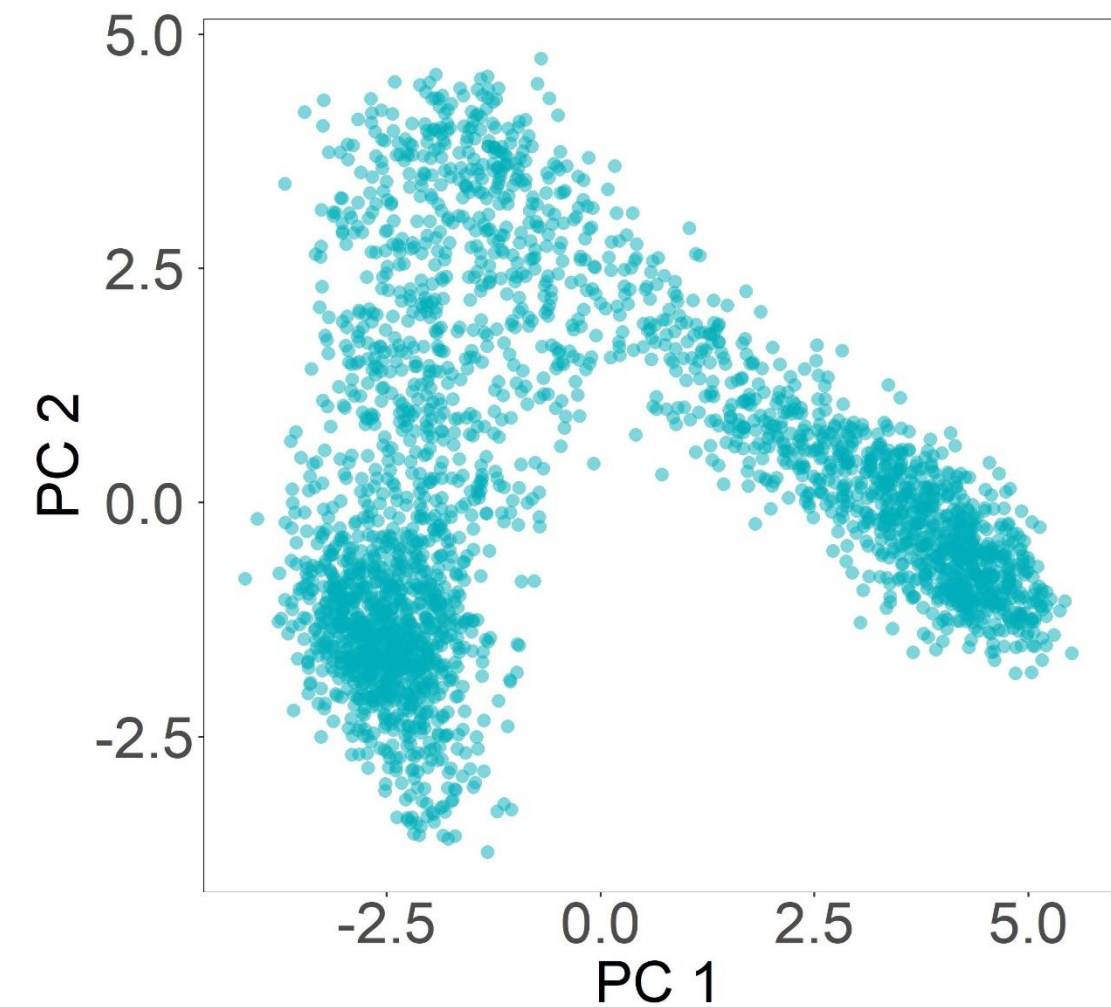
Thousands of cells are unlabeled. Clustering automatically groups cells with similar expression profiles.



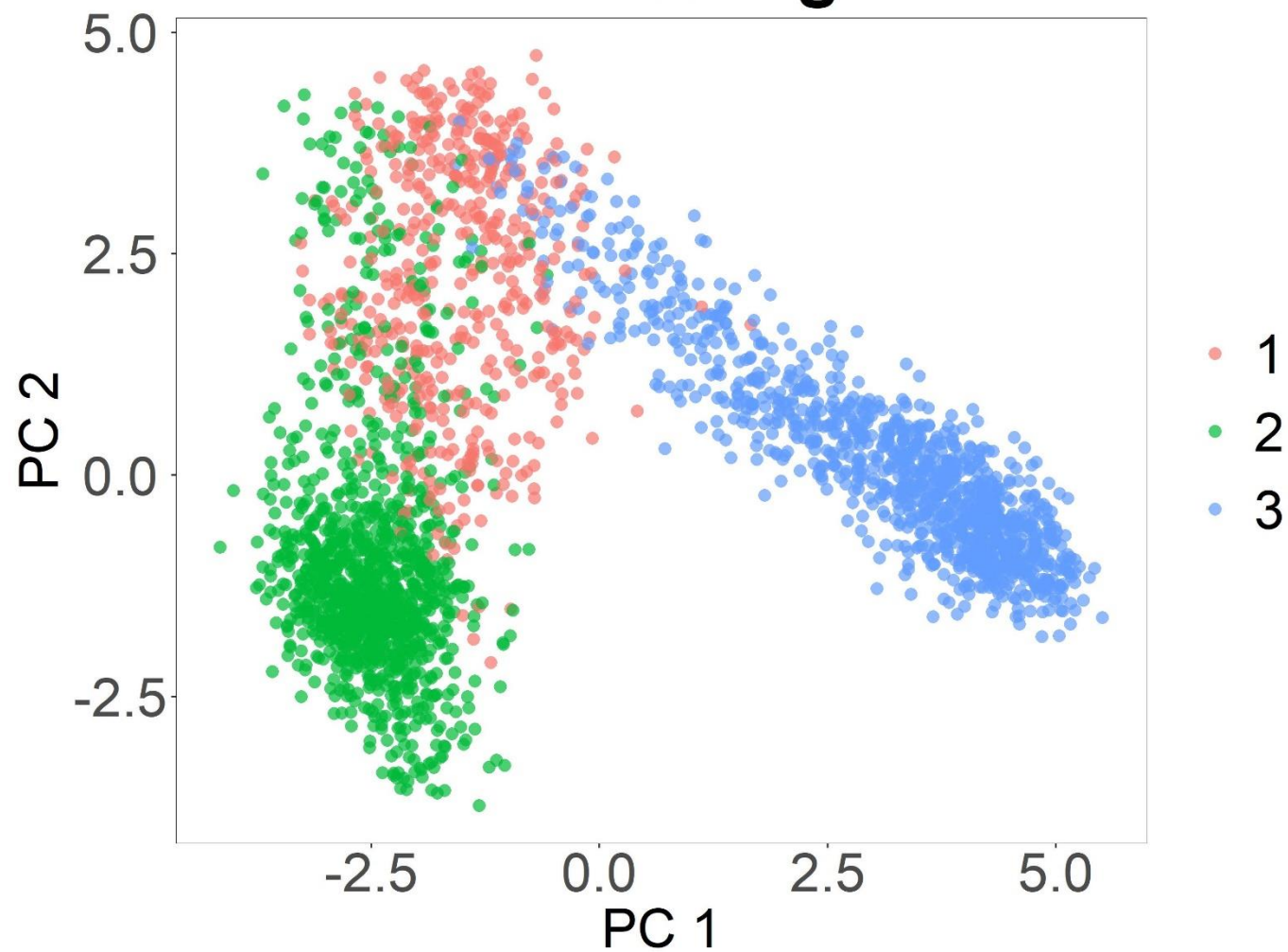
FindNeighbors builds a similarity graph. FindClusters turns dense communities into candidate cell populations.

Clustering

PCA



Clustering



Marker genes

Clusters have numbers, but we still do not know which cell types they represent.



Expression patterns

Cluster 1: Gata2
Cluster 2: Cd34
Cluster 3: Gata1

The next real-data slides use the same genes and lineage example.

Statistical tests find genes enriched in each cluster. Marker patterns provide the evidence for cell-type labels.

Cell-type annotation

Marker patterns are matched to known lineage biology

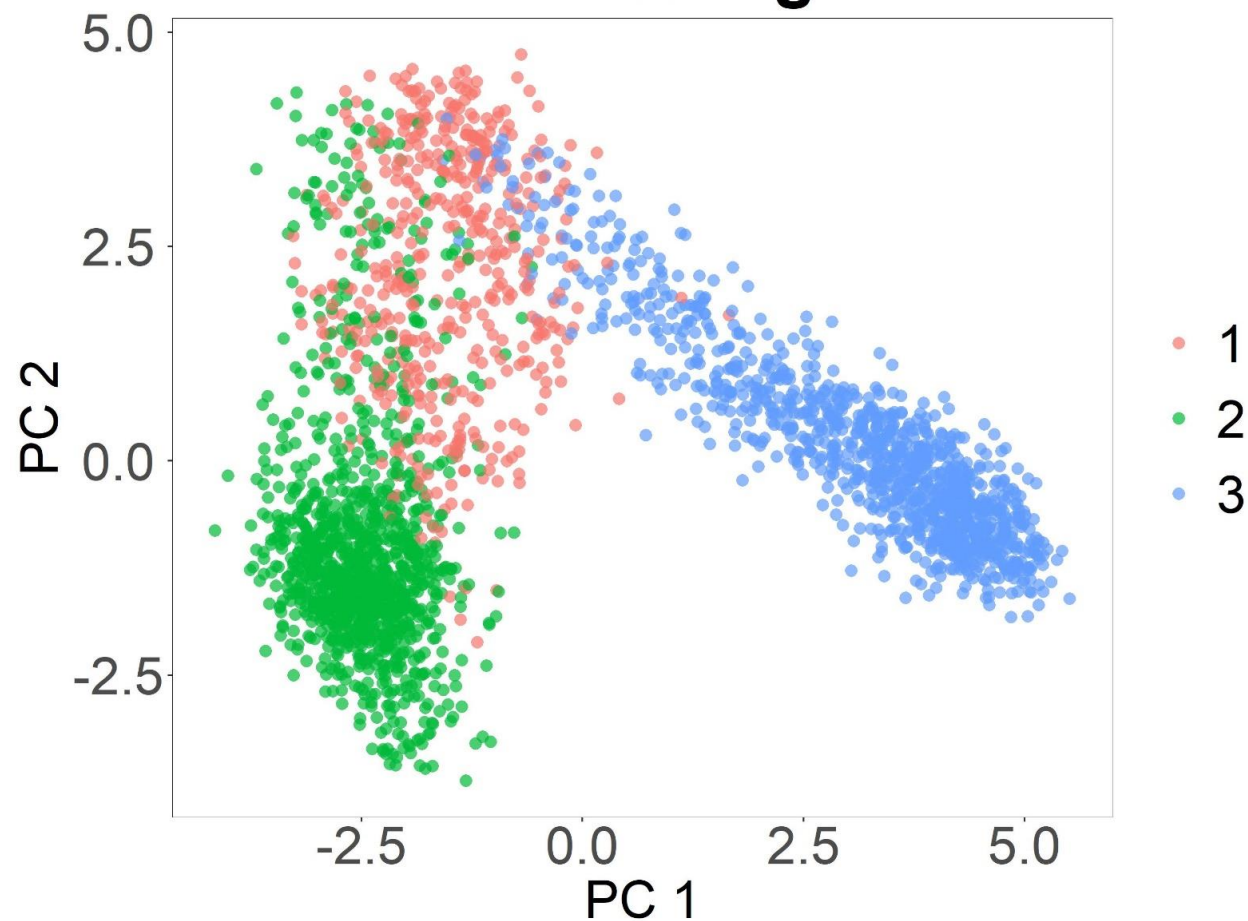
Cluster	Example markers		Label	Interpretation
1	Gata2	→	CMP	common myeloid progenitor
2	Cd34	→	GMP	granulocyte–monocyte progenitor
3	Gata1	→	MEP	megakaryocyte–erythroid progenitor

The algorithm supplies clusters and markers; biological knowledge supplies the names.

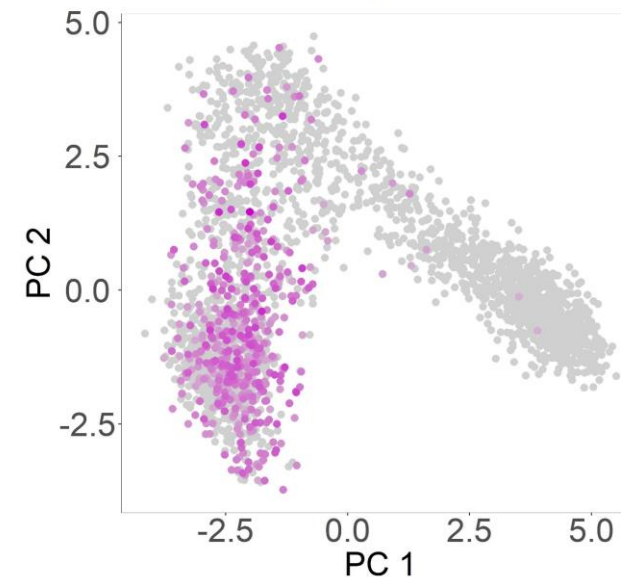
Differential expression analysis

For each gene, a statistical test compares one cluster with the other cells. Adjusted p-values account for testing thousands of genes.

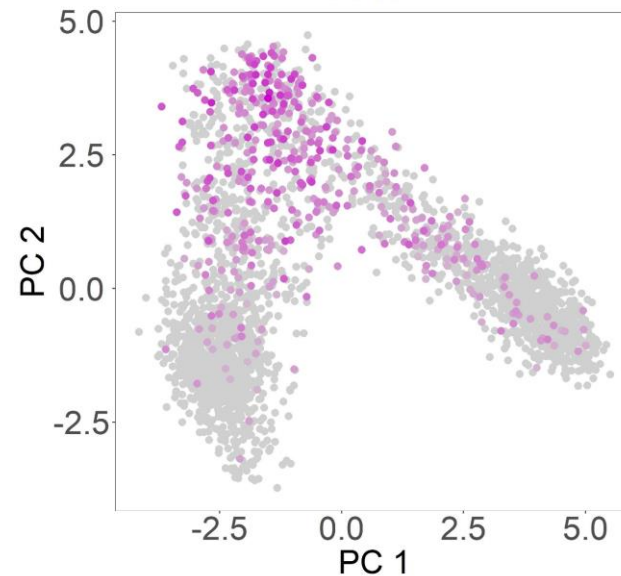
Clustering



Cd34

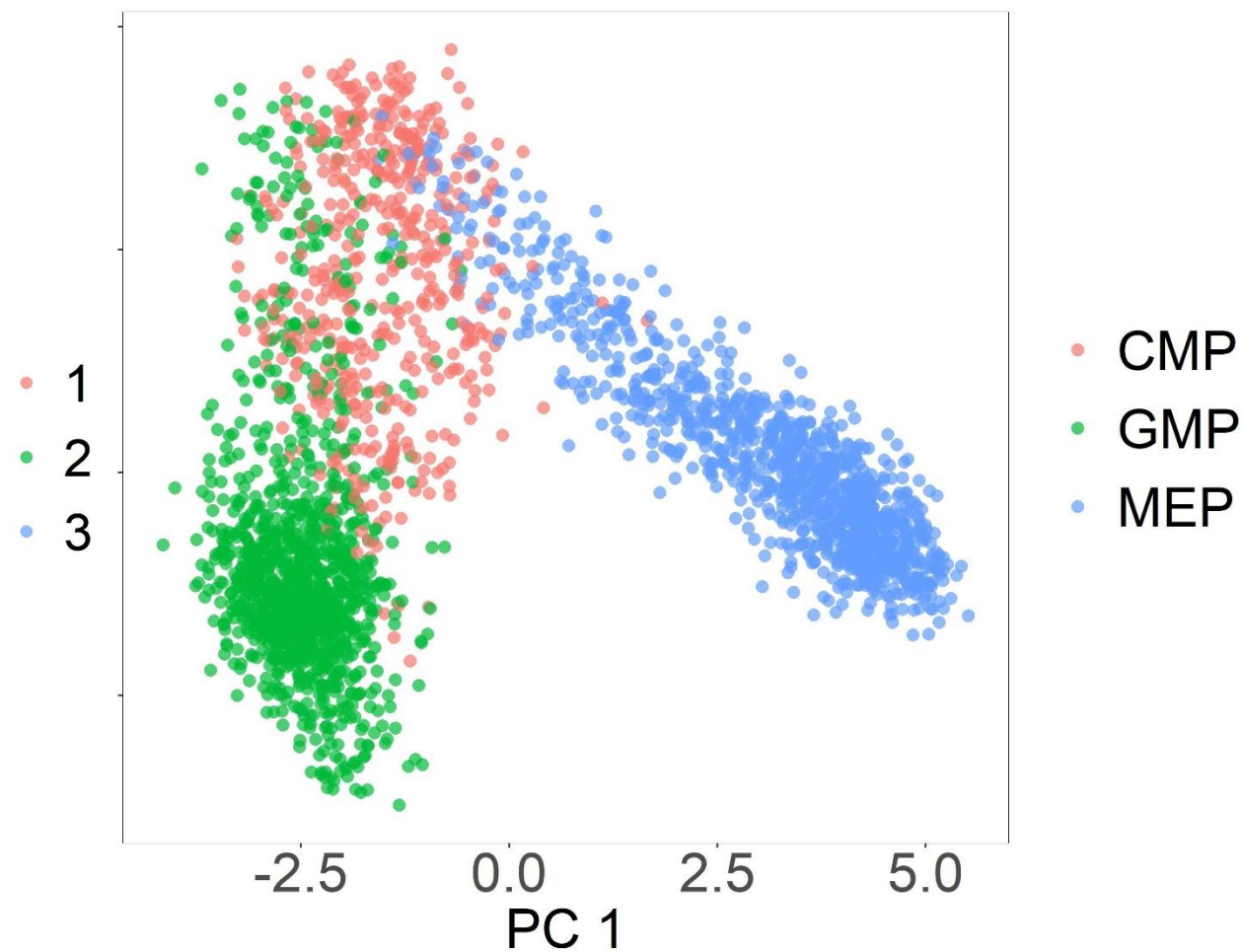
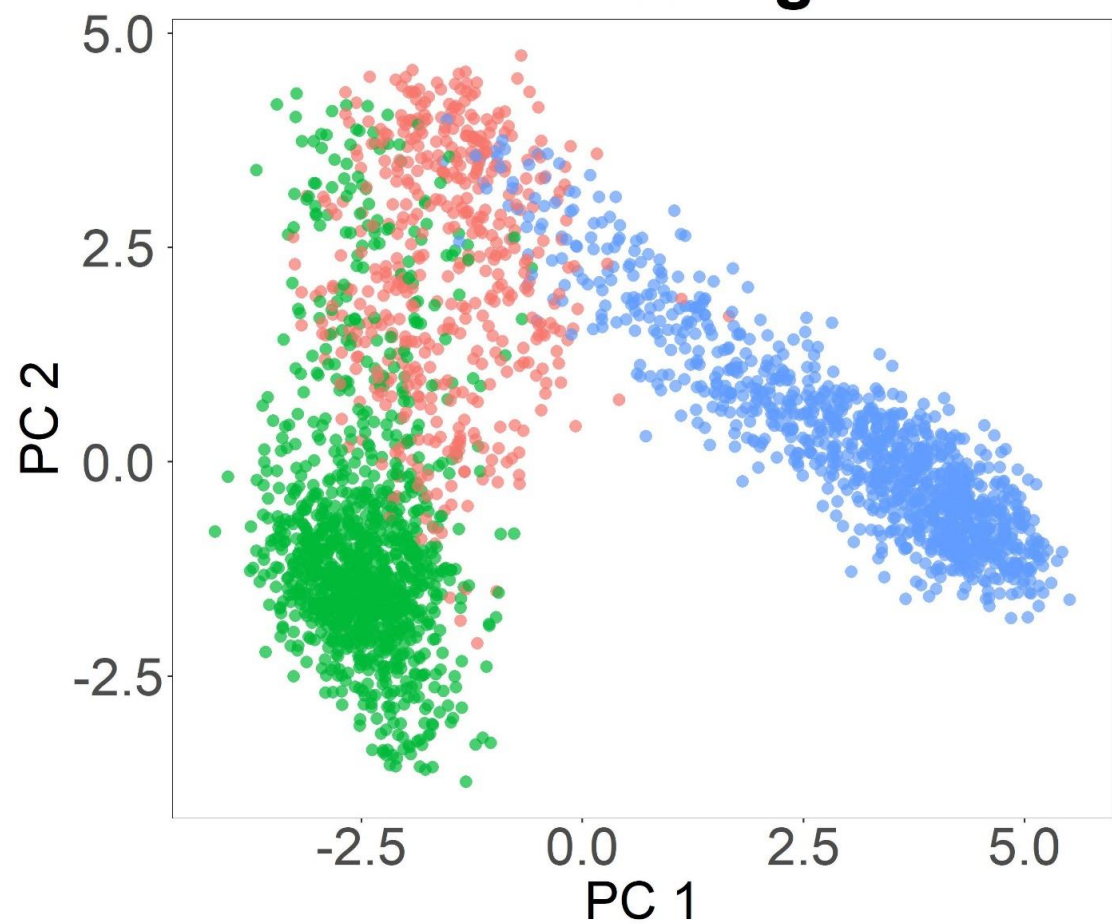


Gata2



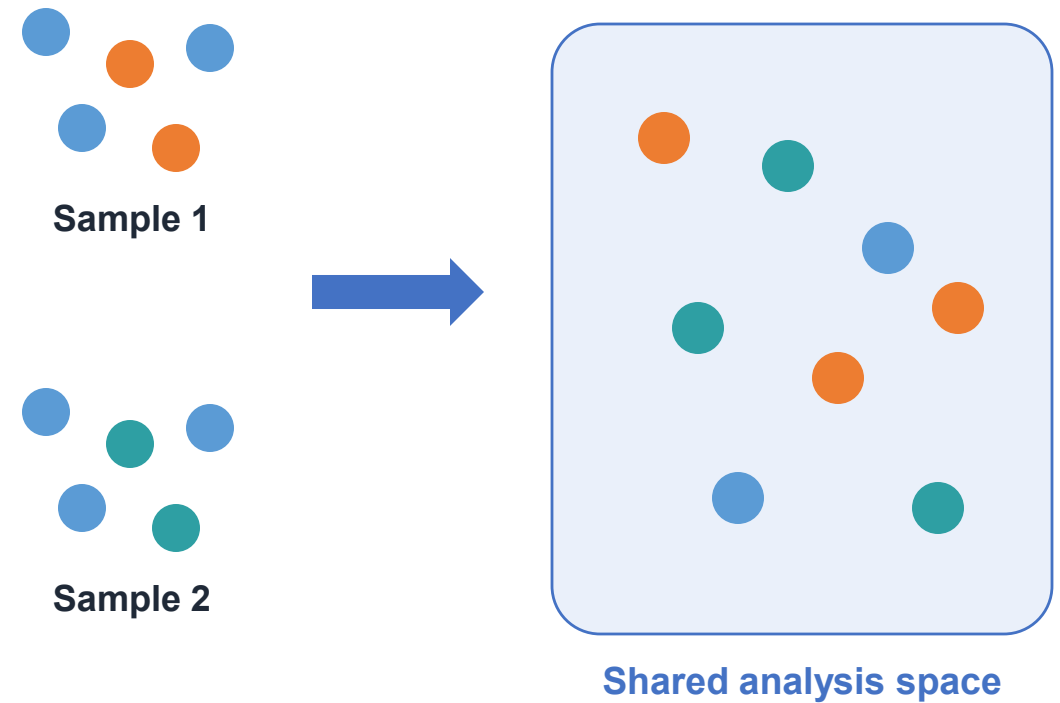
Cluster annotations

Clustering



Integrating multiple samples

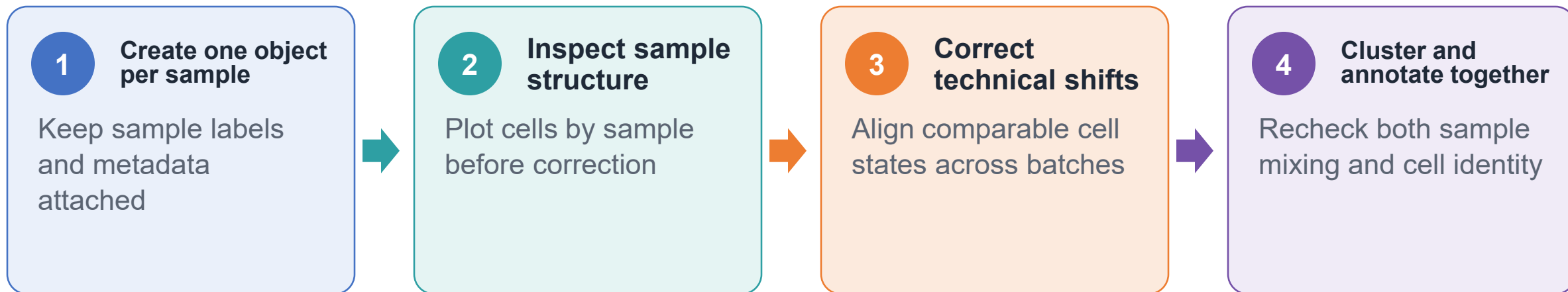
Combine experiments while keeping biological differences and reducing technical shifts.



Goal: align shared cell states without inventing cells or erasing real differences.

Multiple samples, one analysis

Integration is a workflow with checks, not a single button



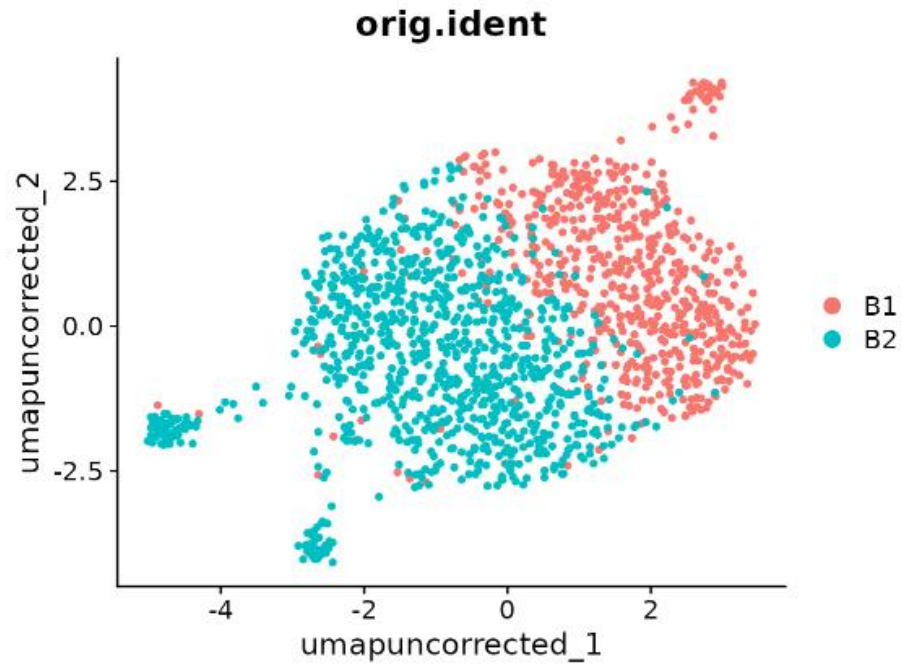
Quality check

Compare plots colored by sample and by cell type before and after correction.

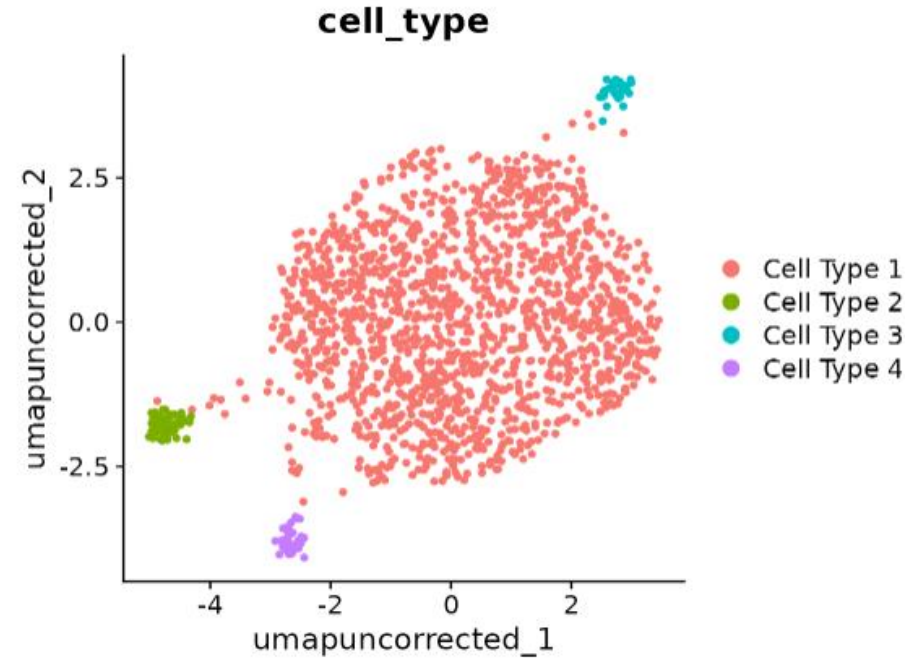
Good integration mixes batches within shared cell states while preserving sample-specific populations.

The batch effect

coloured by sample



coloured by cell type



The same cells, coloured two ways. One cell type splits into two clouds. The split follows the batch, not the biology.

Three integration methods

The workshop compares corrections applied at the same point in the workflow

CCA

Shared variation and anchors

Creates integrated.cca

Harmony

Iterative adjustment in PCA space

Creates harmony

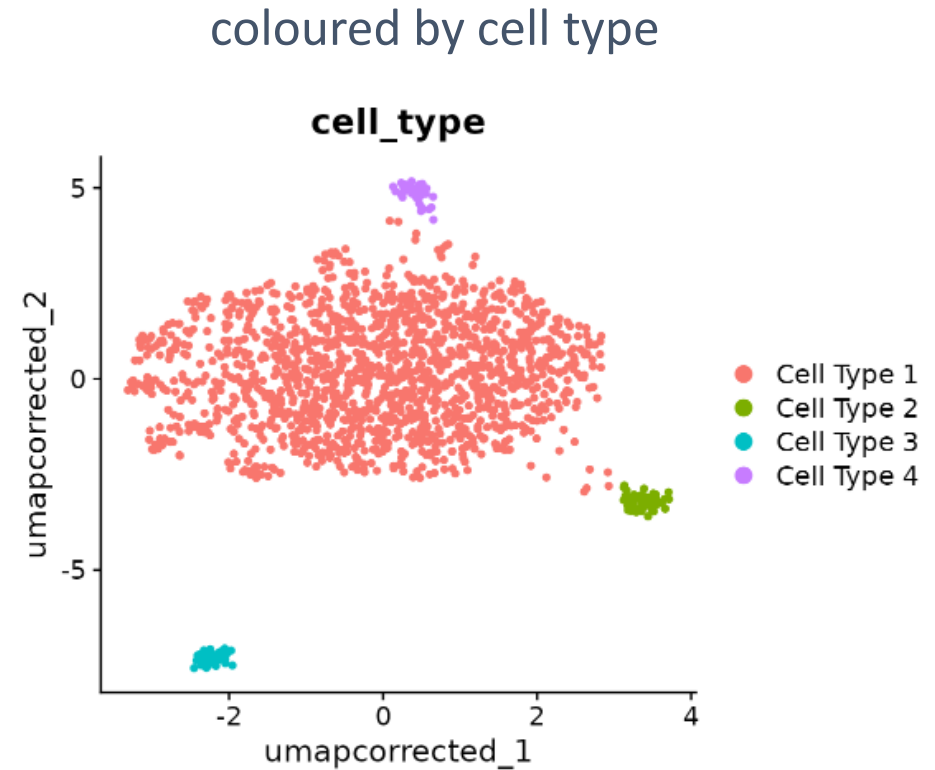
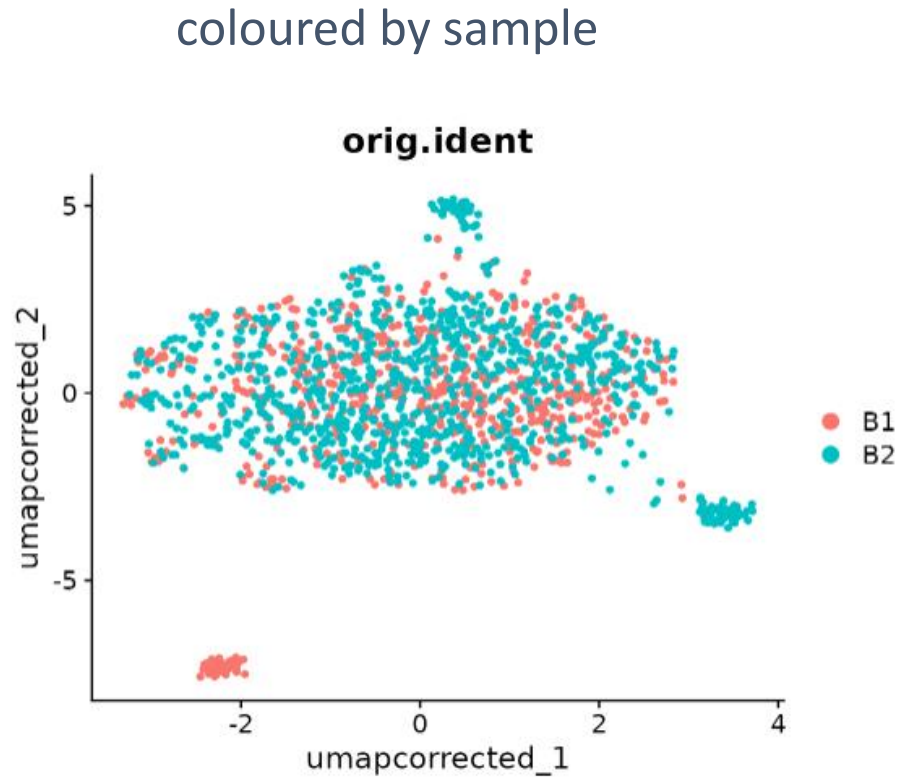
Canek

Mutual nearest neighbors between batches

Creates canek

Many integration methods exist. This workshop focuses on CCA, Harmony, and Canek.

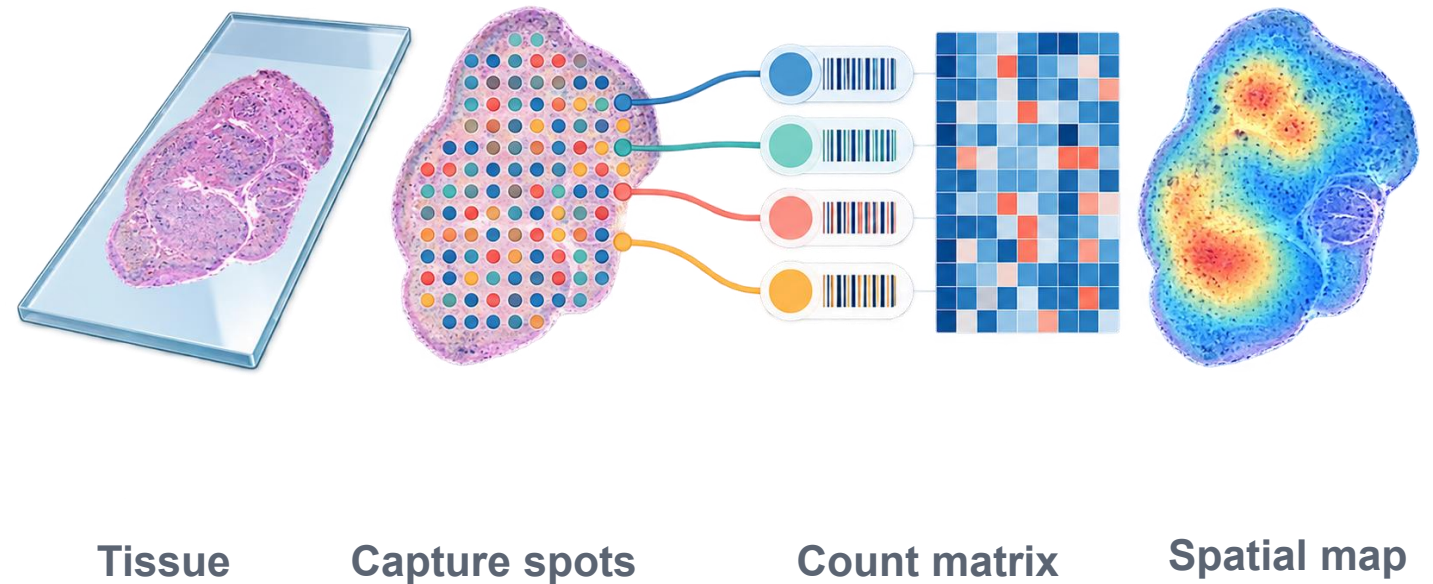
After correction



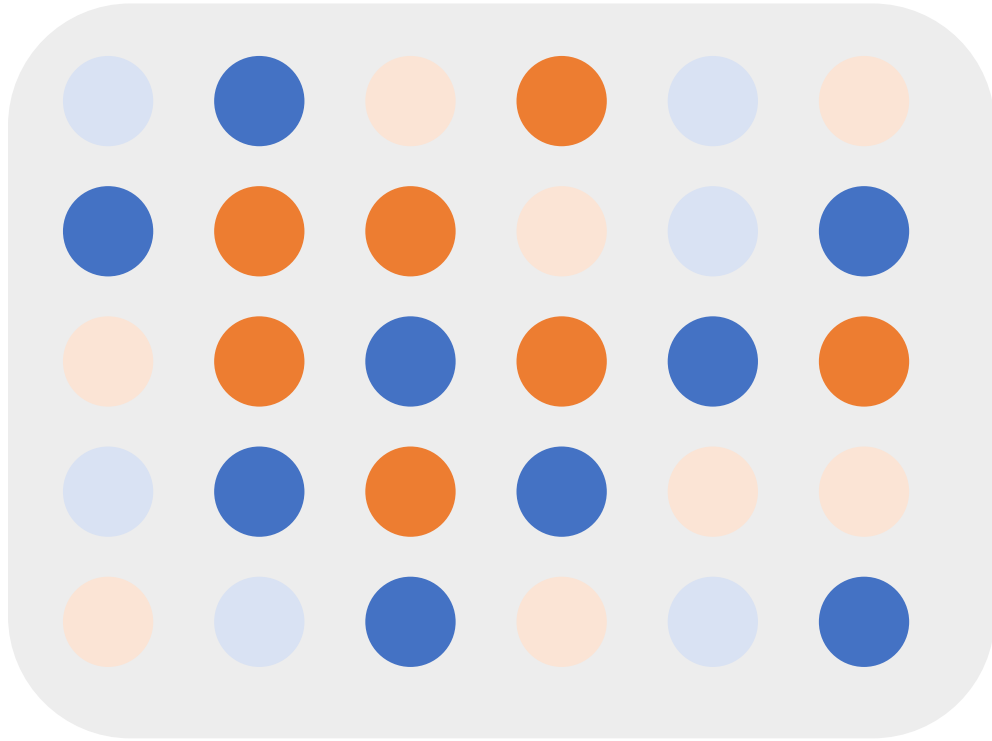
The two samples now overlap. The group that exists in only one sample stays apart, which is correct: integration must not invent cells that were not there.

Spatial transcriptomics

Gene expression remains linked to its position in the tissue, so molecular patterns can be interpreted in anatomical context.



Counts with a position



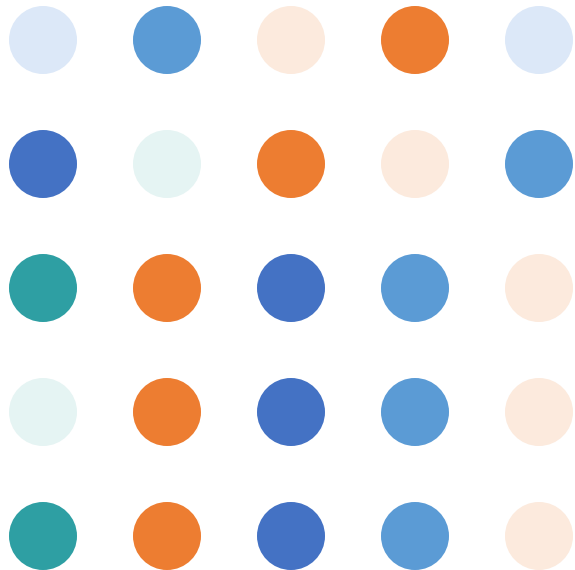
one tissue slide, covered with spots

- The slide is covered with spots in a fixed grid.
- Each spot has its own barcode, so we know which counts came from which spot.
- We also keep the image of the tissue, and the x and y of every spot.
- A spot is not one cell. In Visium, one spot holds about 1 to 10 cells.

Spatial QC and SCTransform

Count depth can follow anatomy, so spatial data require different judgment

Counts across the tissue



Low counts may mark sparse anatomy rather than poor data.

SCTransform

Models the counts expected from sequencing depth, then calculates variance-stabilized residuals



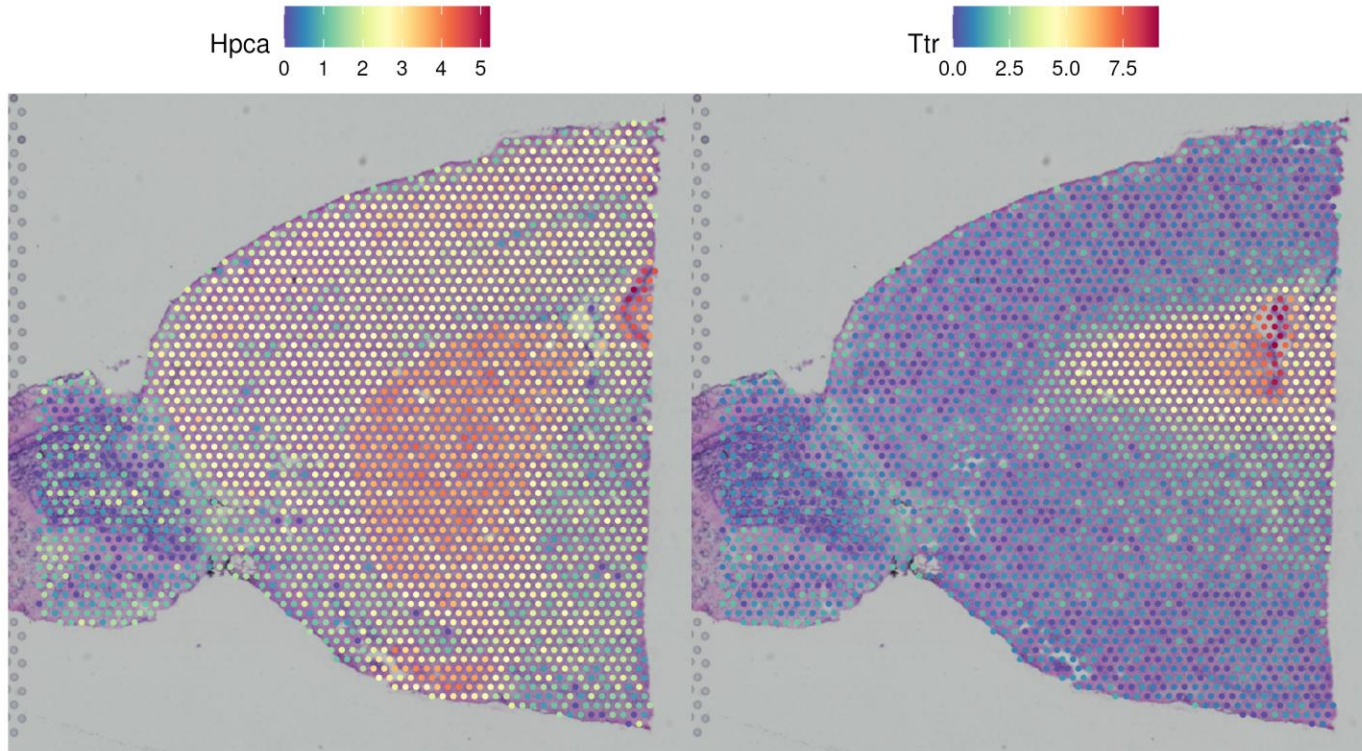
SCT assay

counts + residuals

Raw counts remain in the Spatial assay.

Gene expression in tissue coordinates

SpatialFeaturePlot overlays molecular measurements on the original tissue positions



How to read it

- Each spot keeps its x/y tissue location
- Color shows expression at that spot
- Hpc highlights hippocampal regions
- Ttr highlights choroid plexus

A spatial plot answers both questions: which spots express the gene, and where are those spots in the tissue?

Questions, then hands-on

The final ten minutes are reserved for discussion

**10
min**

Questions

What would you like to revisit before we open the notebooks?

Hands-on sequence

1

Seurat object

2

QC and clustering

3

Integration

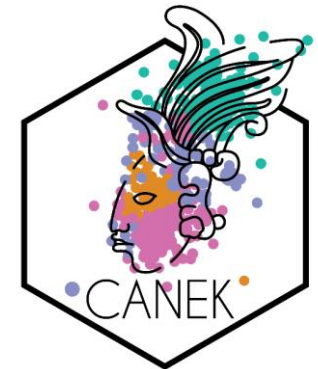
4

Spatial analysis

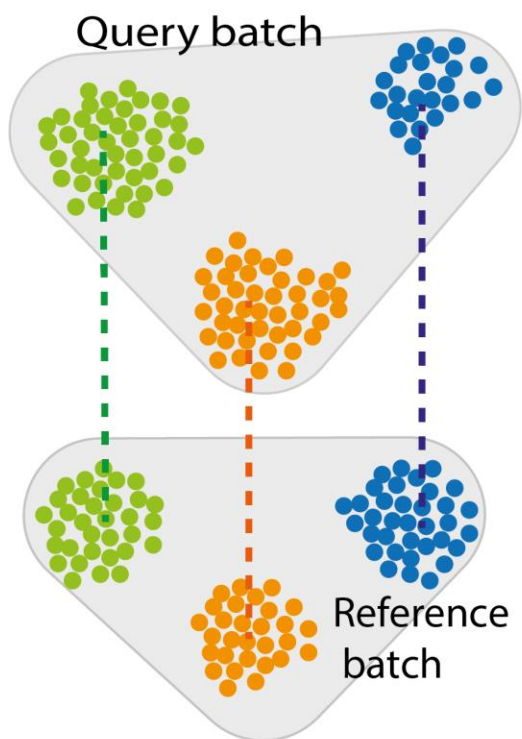
github.com/MartinLoza/CIAD_workshop_sc

Extra reference

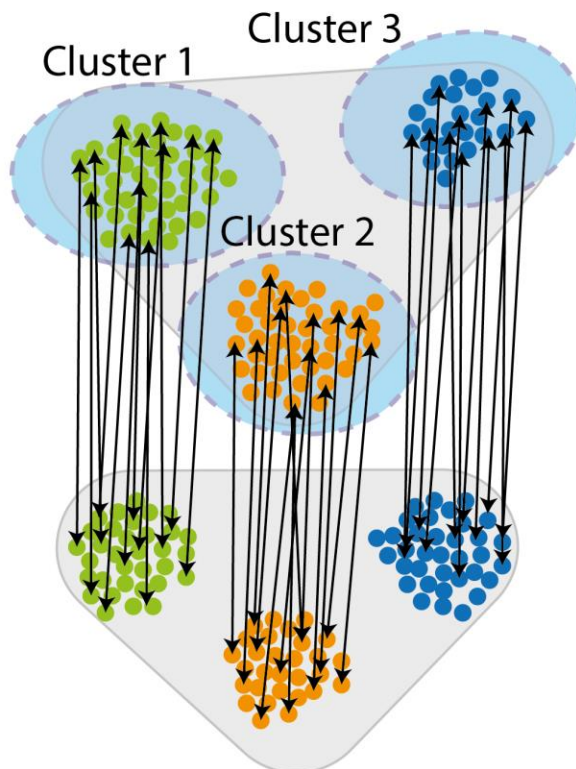
Batch-effect correction algorithm



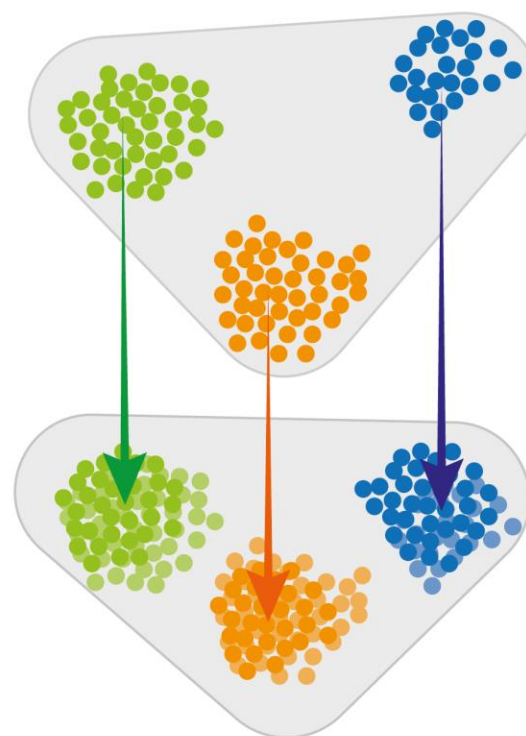
Mostly linear batch-effect



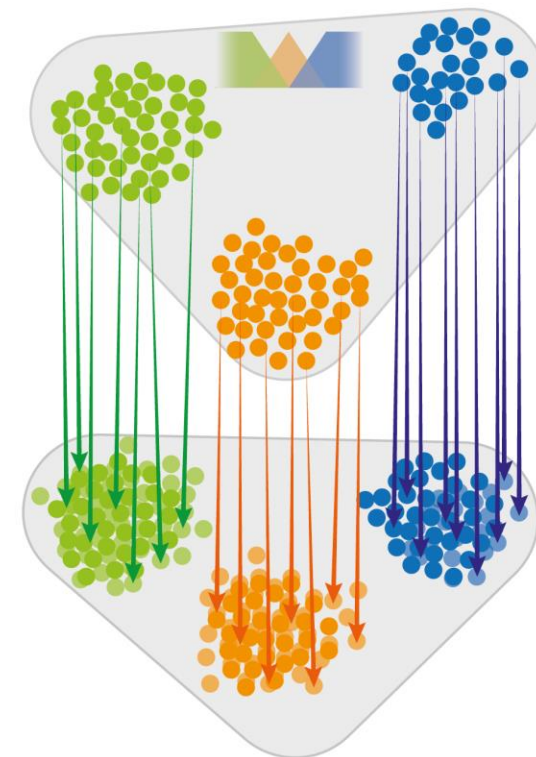
MNNs and Clustering



Linear correction vector



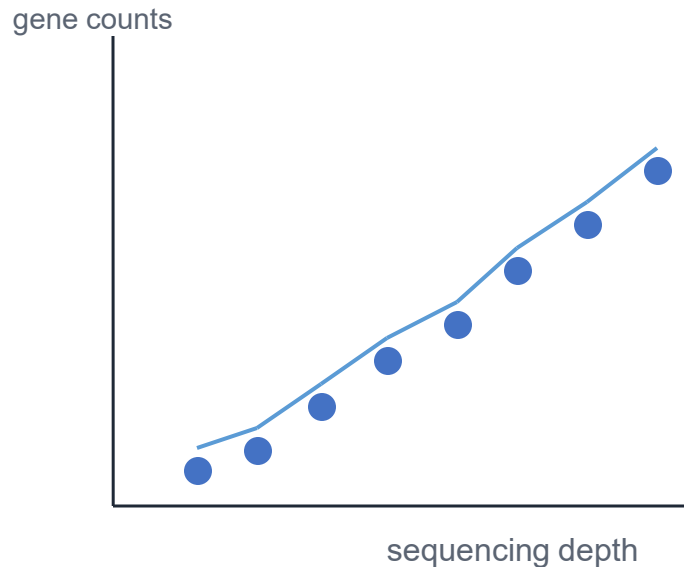
Fuzzy correction



SCTransform in one picture

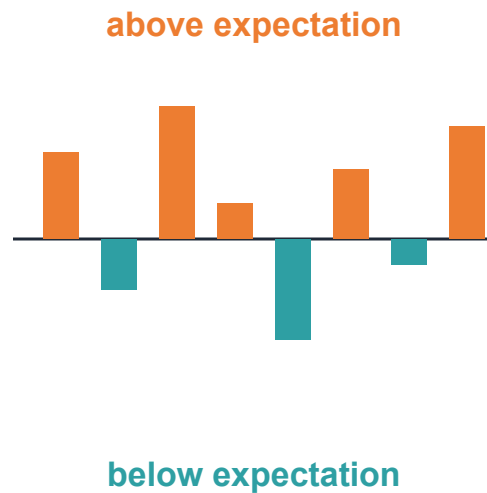
A count model separates depth-related variation from expression above or below expectation

1 Model expected counts



The model learns how counts change with depth.

2 Calculate residuals



Residual = observed count – expected count

3 Store results in the SCT assay

counts	corrected UMI counts
data	log1p corrected counts for visualization
scale.data	Pearson residuals used by PCA

SCTransform replaces NormalizeData, FindVariableFeatures, and ScaleData in the SCT workflow. The main single-cell section uses the standard log-normalization workflow.